

Foot pain

with KARL LANDORF, STEPHEN SIMONS, CHRISTOPHER JORDAN and
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Plantar fasciitis sucks. It feels like you have needles underneath your feet while you're playing.

Joakim Noah, NBA Chicago Bulls, 2013

Injuries to the foot, both acute and chronic, can be extremely challenging for the clinician, largely due to the complex anatomy (Figs 43.1 and 43.2). If the foot is considered in its three distinct regions (Fig. 43.1)—the rearfoot (calcaneus and talus), the midfoot (the navicular proximally and medially, the cuboid laterally, and the cuneiforms distally and medially), and the forefoot (the metatarsals and phalanges)—the bony anatomy is greatly simplified. Soft tissue anatomy can then be superimposed on the regional division of the foot (Figs 43.2c-e).

In this chapter we will consider the three different clinical presentations:

- rearfoot pain (mainly the heel)
- midfoot pain
- forefoot pain.

REARFOOT PAIN

The most common cause of rearfoot (inferior heel) pain is plantar fasciopathy/fasciitis, often mistakenly referred

to as 'heel spur'. This condition occurs mainly in runners and the older adults, and may be associated with a biomechanical abnormality, such as excessive pronation or supination.¹ Another common cause of heel pain is fat pad syndrome or fat pad contusion. This is also known as a 'bruised heel' or a 'stone bruise'.

Less common causes of heel pain are stress fracture of the calcaneus and conditions that refer pain to this area such as tarsal tunnel syndrome or medial calcaneal nerve entrapment (Chapter 42). Causes of rearfoot pain are listed in Table 43.1.

History

The pain of plantar fasciopathy is usually of insidious onset, whereas fat pad damage may occur either as a result of a single traumatic episode (e.g. jumping from a height onto the heel) or from repeated heel strike (e.g. on hard surfaces with inadequate heel support). Plantar fasciopathy pain is typically worse in the morning, improves with exercise at first and is aggravated by standing.

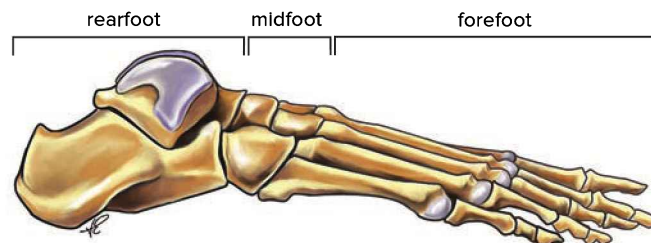
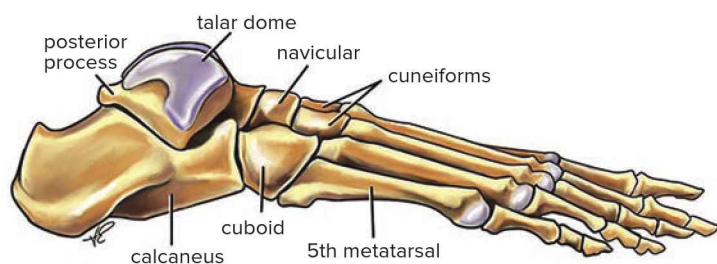
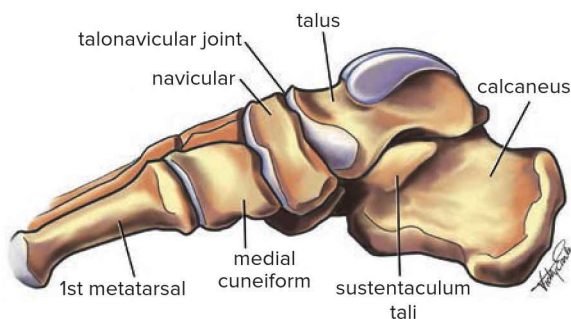


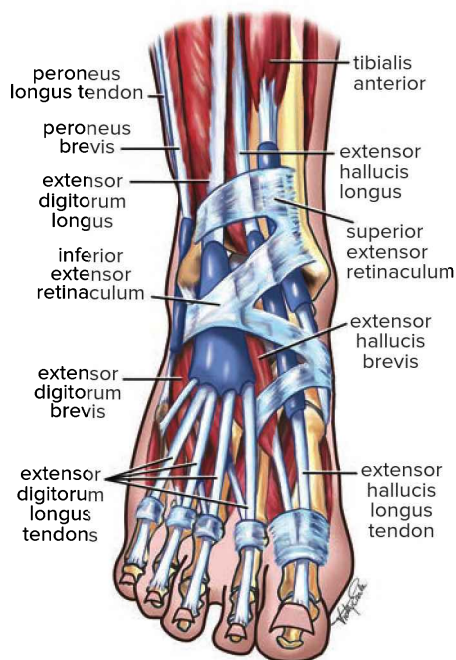
Figure 43.1 The regions of the foot: rearfoot, midfoot and forefoot



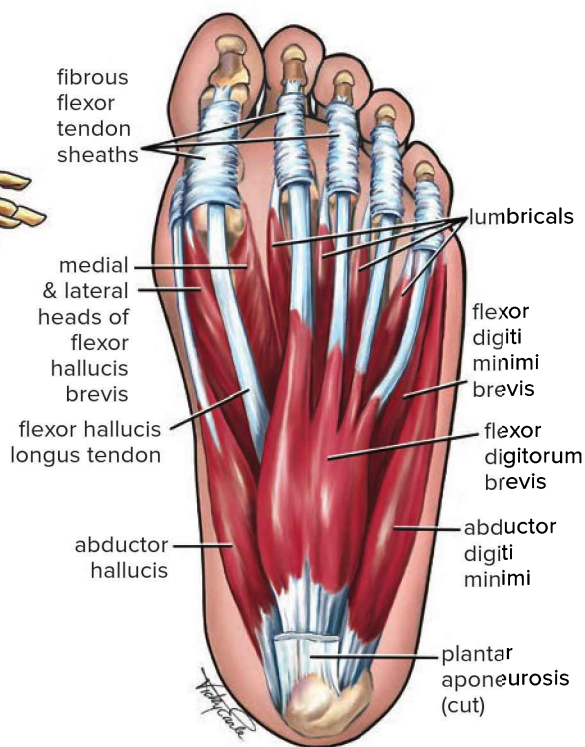
(a)



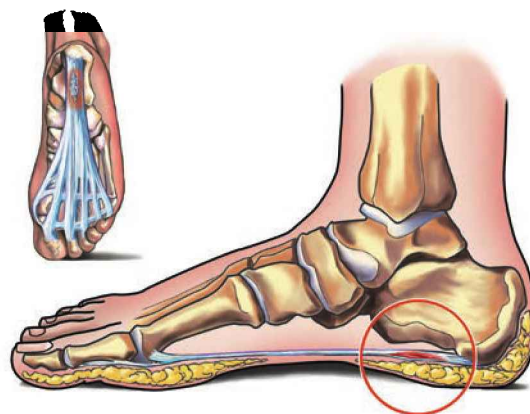
(b)



(c)



(d)



(e)

Figure 43.2 Anatomy of the foot (a) Lateral view of the bones of the foot (b) Medial view of the bones of the foot (c) Dorsal view of the soft tissues of the foot (d) Plantar view of the soft tissues of the foot: first layer (e) Plantar fascia

Table 43.1 Causes of rearfoot and inferior heel pain

Common	Less common	Not to be missed
Plantar fasciopathy (fasciitis) Fat pad contusion	Calcaneal stress fracture Medial calcaneal nerve entrapment (Chapter 42) Lateral plantar nerve (Baxter's nerve) entrapment Tarsal tunnel syndrome (Chapter 42) Talar stress fracture (Chapter 42) Retrocalcaneal bursitis (Chapter 40)	Spondyloarthropathies Osteoid osteoma Complex regional pain syndrome type 1 (after knee or ankle injury)

Examination

Examination of the rearfoot is shown in Figure 43.3. The windlass (or Jack's) test (passive dorsiflexion of the first metatarsophalangeal joint that provokes pain) is a quick and highly specific test for the plantar fascia (Fig. 43.3c).²

Biomechanical assessment is an important component of the examination and should include ankle, subtalar and

midtarsal joint range of motion. Also assess function of the forefoot and first metatarsophalangeal (MTP) joint range of motion. Examine footwear, looking for asymmetrical wear of the sole which may indicate differences in function between left and right sides.

Investigations

Radiographs only contribute to the clinical work-up of rearfoot pain in a small proportion of cases. It may reveal a calcaneal spur which is associated with increased risk of heel pain. Plain radiograph is generally normal in stress

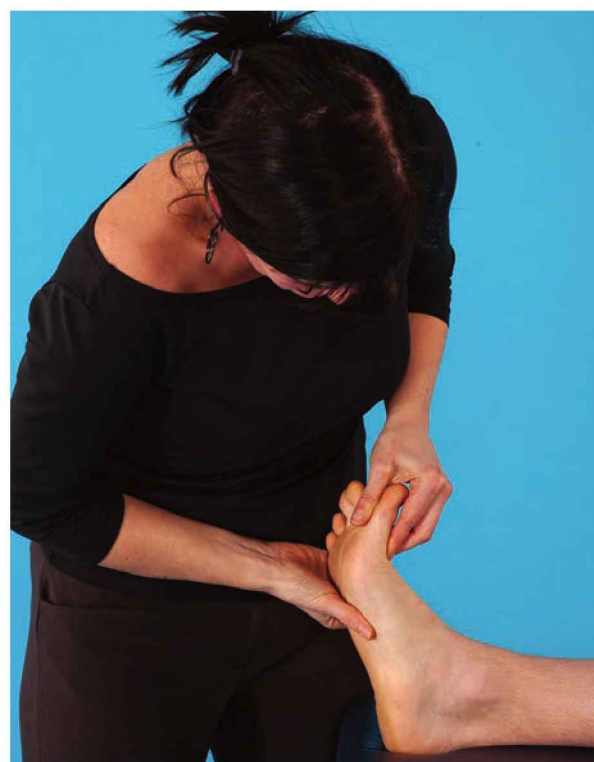


(a)



(b)

Figure 43.3 Examination of the rearfoot (a) Palpation—medial process of calcaneal tuberosity. Palpate plantar fascia attachment (b) Palpation—heel fat pad



(c)

Figure 43.3 (cont.) (c) The windlass (or Jack's) test—passive dorsiflexion of the first metatarsophalangeal joint

fractures of the calcaneus, but if the injury has been present for many weeks, there may be a line of sclerosis (increased opacity).

Magnetic resonance imaging (MRI) is the investigation of choice for stress fracture. MRI and ultrasound can each confirm the presence of plantar fasciopathy. MRI reveals increased signal intensity and thickening at the attachment of the plantar fascia to the calcaneus at the medial calcaneal tuberosity, often with oedema in the

adjacent bone. Ultrasound reveals a characteristic region of hypoechogenicity.

Patient-reported outcome measures

There are a number of patient-reported outcome measures (PROMs) which are relevant for patients with foot pain. Combined Foot and Ankle PROMs have been described in Chapter 42. The foot specific PROMs are shown in Table 43.2.

Table 43.2 Patient-reported outcome measures for the foot

Foot Function Index (FFI)
• Self-administered questionnaire evaluating pain, disability and activity restrictions • 23 items (9 on pain, 9 on disability and 5 on activity limitation) • Score range 0–100 for all subscales (100 = maximal impairment)³ • Test-retest reliability: - Total score, ICC = 0.87³ - Pain subscale, ICC = 0.70³ - Disability subscale, ICC = 0.84³ - Activity limitation subscale, ICC = 0.81³ • Construct validity: - Pain subscale: SF-36: bodily pain subscale: –0.61; physical component summary scale: –0.45⁴ - Disability subscale: SF-36: physical component summary scale: –0.67; physical functioning: –0.69; role-physical: –0.62; bodily pain: –0.64⁴ - Activity limitation subscale: physical component summary scale: –0.53; physical functioning: –0.55; role-physical: –0.64; social functioning: –0.57⁴ - Total score: SD-36; physical component summary scale: –0.64;⁵ AOFAS, Hallux Metatarsophalangeal-Interphalangeal Scale: –0.81 • Developed and validated in American English³ and there is a revised-FFI version available with 34 items⁶ • The original FFI was cross-culturally validated to Danish,⁷ French,⁸ German,⁹ Italian,^{10, 11} Korean,¹² Spanish¹³ and Taiwan Chinese¹⁴
Rowan Foot Pain Assessment Questionnaire
• Self-administered questionnaire evaluating pain in sensory, affective and cognitive dimensions • 36 items (16 on the sensory, 10 on the affective and 10 on the cognitive dimension) • Score ranges from 0–5 for each subscale • Test-retest reliability: - Sensory subscale: Spearmans ρ = 0.88¹⁵ - Affective subscale: Spearmans ρ = 0.93¹⁵ - Cognitive subscale: Spearmans ρ = 0.82¹⁵ • Construct validity: - Sensory subscale: FFI_{pain} = 0.88¹⁵ - Affective subscale: FFI_{pain} = 0.69¹⁵ - Cognitive subscale: FFI_{pain} = 0.70¹⁵ • Developed and validated in English
Rowan Foot Pain Assessment Questionnaire
• Self-administered questionnaire evaluating foot pain, foot function, footwear and general foot health • 13 items (4 items on foot pain, 4 items on foot function, 3 items on footwear, 2 items on general foot health) • Score range 0–100 (100 = no health problem)

- Test-retest reliability:
 - Foot pain: ICC = 0.86
 - Foot function: ICC = 0.92
 - Footwear: ICC = 0.74
 - General foot health: ICC = 0.78
- MCID:
 - Foot pain: 13¹⁶
 - Foot function: 7¹⁶
 - Footwear: 2¹⁶
 - General foot health: 0¹⁶
- Developed and validated¹⁷ in Australian English; cross-culturally validated to Brazilian Portuguese¹⁸ and Spanish¹⁹

Rewan Foot Pain Assessment Questionnaire

- Self-administered questionnaire evaluating walking, hygiene, nail care, foot pain, worry/concern QoL and general foot status
- 6 items + 1 visual analogue scale (VAS) regarding foot status
- 6 items score range: VAS Score range 0–100 (100 = best possible foot health)
- Construct validity:
 - Walking: EQ-5D^{mobility}: 0.54; EQ-5D^{usual activities}: 0.45; EQ-5D^{pain/discomfort}: 0.49²⁰
 - Hygiene: EQ-5D^{self-care}: 0.58²⁰
 - Nail care: EQ-5D^{self-care}: 0.37²⁰
 - Foot pain: EQ-5D^{pain/discomfort}: 0.40²⁰
 - Worry/concern: EQ-5D^{anxiety/depression}: 0.21²⁰
 - Quality of life: EQ-5D^{mobility}: 0.54, EQ-5D^{usual activities}: 0.34²⁰
- Developed and validated in British English

Plantar fasciopathy (formerly called ‘plantar fasciitis’)

Plantar fasciopathy (also known as plantar heel pain or plantar fasciitis/fasciosis) is an overuse condition of the plantar fascia at its attachment to the calcaneus.

PRACTICE PEARL

The histology is of degenerative changes at the plantar fascia enthesis, including a deterioration of collagen fibres, increased secretion of ground substance proteins, focal areas of fibroblast proliferation and increased vascularity.^{21–23}

The plantar fascia consists of type 1 collagen and has common tendinopathy traits, hence we recommend using the term ‘fasciopathy’ rather than the former ‘plantar fasciitis’. Plantar fasciopathy is the most common cause of plantar heel pain, which has an estimated prevalence of 4–7%.^{24, 25} It commonly affects very physically active people (e.g. runners) or people with high amounts of standing occupational work (who may also have a high BMI). Through this section we try to differentiate between the two groups and discuss how treatment strategies might be tailored to each overarching group.

Risk factors

Only one true prospective study has investigated risk factors for plantar fasciopathy. The study was conducted in a cohort of 166 runners followed over 5 years.^{26, 27} The study found the overall incidence of plantar fasciopathy over the 5-year period to be 31%. Four risk factors that were significantly associated with an increased risk of plantar fasciopathy: varus knee alignment (OR 5.63); use of spiked athletic shoes (OR 5.49); cavus foot posture (OR 5.52); and greater number of days of practice per week (OR 2.59). High BMI was not a risk factor for developing plantar fasciopathy in this cohort of runners with low BMIs. This suggests that high training volumes that increase load or vertical force on the plantar fascia may increase a runner’s risk of plantar fasciopathy which fits well with the belief that it is an overuse type injury.

Weight and BMI

One of the most common findings is that patients with plantar fasciopathy have a high BMI, although many of the studies that contribute to this evidence are cross-sectional.²⁷ Increased mechanical load due to higher BMI is a plausible source of increased plantar fascial stress. However, an important point is that high BMI was not a risk factor in the study in runners,²⁶ which may suggest that a high BMI increases risk in more sedentary people and in people with occupational work that requires many hours of standing, particularly on harder surfaces.

Alignment of the foot

Cross-sectional studies that have investigated the association between foot type and plantar fasciopathy provide conflicting evidence.²⁷ One prospective study found that runners with a cavus foot have a higher risk of plantar fasciopathy, but cross-sectional research fails to identify a strong link between foot type and plantar fasciopathy.²⁶ It may be that in some patient types (e.g. those with standing work and high BMI), foot type is important. Nevertheless, a recent systematic review found that overall, the evidence supporting associations for foot and ankle range of motion and strength, kinematic and kinetic factors, and foot posture was either inconsistent or inconclusive.²⁷

Training volume

Plantar fasciopathy in the younger, athletic population is associated with activities that cause a dose-dependent repetitive stress to the plantar fascia, such as running.^{26, 28} Accordingly, one of the most important management strategies in these cases may be to adjust training volume and educate the patient with plantar fasciopathy to adjust training load for a period.

Standing work

People who spend the majority of their work day on hard surfaces and on their feet may be at higher risk for developing plantar fasciopathy.²⁷ This has only been shown in cross-sectional data, but it fits well with the proposition that high mechanical loads may increase an individual's risk of developing plantar fasciopathy. This may have implications for occupational interventions (more in the section on treatment).

Flexibility and strength

Plantar fasciopathy may be associated with decreased flexibility of the ankle joint and hamstrings, as well as decreased plantar flexion peak torque. Likewise, patients with plantar fasciopathy may have significantly lower toe flexor strength. It is unknown if this is a cause or an effect of longstanding plantar fasciopathy.²⁷ However, these findings may be important as treatment may need to address these strength and flexibility deficits.

Plantar fasciopathy may be associated with tightness in the proximal myofascial structures, especially the calf, hamstring and gluteal regions. Tightness in these muscle groups can alter foot biomechanics.²⁹

Clinical features

The pain related to plantar fasciopathy is usually of gradual onset and felt classically on the inferior medial aspect of the heel. Initially, it is worse in the morning when getting out of bed (referred to as 'first-step pain') and decreases

with activity, only to return with an ache post-activity. Periods of inactivity during the day are generally followed by an increase in pain as activity is recommenced. As the condition becomes more severe, the pain may be present when standing and worsen with activity. Some patients may also experience atypical pain when non weight-bearing, for example, when they go to bed at night.

Examination reveals pain or tenderness along the medial tuberosity of the calcaneus, and this may extend some centimetres along the medial and central components of the plantar fascia. Stretching the plantar fascia via dorsiflexion of the hallux, such as during the windlass test, may reproduce pain and may assist with palpation of the plantar fascia (Fig. 43.3c).

Investigations

If the patient has the classic history and examination findings, and there is no other suggestion of a 'red flag' (Chapter 7 and Table 43.1), then medical imaging or other investigations are not indicated. Further investigation is recommended however, if pain is particularly persistent without responding to treatment (e.g. >1 month, raising suspicion of other pathologies; Table 43.1) or if the nature of the pain is atypical for plantar fasciopathy (acute onset of pain, persistent pain even at rest, systemic symptoms, night pain). Blood tests may be warranted to rule out inflammatory arthritis, which can cause symptoms in the heels, although frequently symptoms will be more widespread. The primary imaging modalities that are useful include plain radiographs, ultrasound and MRI.

Plain radiographs, primarily the lateral view (Fig. 43.4a) are mainly used to investigate for the presence of a plantar calcaneal spur. The radiograph appearance of 'calcaneal spur' is eight times more likely in patients with plantar fasciopathy compared to those without plantar fasciopathy.³⁰ However, calcaneal spurs are also common in people without heel pain, and the presence of a spur does not indicate that this is the cause of pain. Plain radiographs of both feet are useful as they may show spurs bilaterally and allow comparison between feet in the situation of unilateral symptoms. Fractured spurs have also been reported, and although the findings may be subtle, plain radiographs can be used in the first instance to investigate whether a spur has, indeed, fractured.³¹ Lastly, plain radiographs can be used to rule out other bone pathology (e.g. bone cysts).

Ultrasound is particularly useful for detecting pathological changes within the plantar fascia itself. Ultrasound may reveal thickening, hypoechogenicity and heterogeneity, delamination and tears (Fig. 43.4b). A systematic review and meta-analysis found that patients with plantar fasciopathy are 105 times more likely to have a fascia greater than 4.0 mm thick and 204 times

more likely to have hypoechogenicity compared to controls without fasciopathy.³⁰ Ultrasound can also be used to assess for other soft tissue pathologies associated with plantar fasciopathy, such as tissue change within the plantar fat pad, myopathies (e.g. of the deeper intrinsic musculature) and neuroma (e.g. of the medial calcaneal nerve).

MRI is the investigation of choice if there are atypical symptoms and if a wider selection of tissues (e.g. bone) are of interest. Similar to ultrasound, MRI will detect thickening of the plantar fascia, as well as heterogeneity, delamination and tears (Fig. 43.4c). MRI signs of plantar fasciopathy include hyperintensity within and around the plantar fascia, and delamination and tears within the plantar fascia. MRI is also useful for determining acute versus more chronic pathological change via the intensity of the signal on sagittal fat-suppressed short tau inversion recovery (STIR)/T2 images (lower signal signifying more chronic pathology). In addition, there is some evidence to suggest that a subsection of patients with plantar

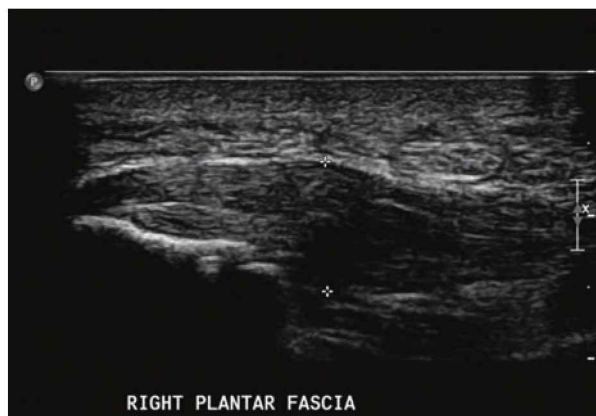
fasciopathy have bone marrow oedema on MRI, and this may be associated with atypical symptoms such as night pain.³² MRI can also identify stress fractures of the calcaneus. Finally, MRI is particularly useful for detecting other soft tissue pathologies that may be associated with a painful heel, including soft tissue tumours, medial tendinopathy and plantar fat pad pathology.

Treatment

Treatment is often multi-faceted and generally starts with conservative interventions, progressing to more invasive interventions if conservative care fails to improve the condition. Treatment can be thought of in terms of three



(a)



(b)



(c)

Figure 43.4 (a) Lateral plain radiograph highlighting a large plantar calcaneal spur (b) Ultrasound scan highlighting a thickened plantar fascia (0.902 cm), heterogeneity and hypoechogenicity within the plantar fascia (deep hypoechogenicity representative of a tear within the fascia) (c) MRI of plantar fasciopathy. Sagittal fat-suppressed T2-weighted MRI demonstrates thickening of the plantar fascia and hyperintense signal in the adjacent perifascial soft tissues (arrows). There is bone marrow oedema in the calcaneal attachment (arrowhead)³³

(c) IMAGING IN SPORTS-SPECIFIC MUSCULOSKELETAL INJURIES 2016, P. 528. ALI GUERMAZI, FRANK W ROEMER, MICHEL D CREMA. © SPRINGER INTERNATIONAL PUBLISHING SWITZERLAND 2016. WITH PERMISSION OF SPRINGER

tiers, which essentially equate to short-, medium- and long-term options.³⁴ The tiers also relate to the complexity, cost and invasiveness of the intervention. The following interventions are listed, taking into account evidence from two systematic reviews, unless otherwise indicated.^{35, 36} The evidence is summarised in Table 43.3.

First tier/short term

Minimise the aggravating activity and educate the patient about the potential causes of their heel pain. If the patient is a runner, it is essential that you also discuss training loads, and regression and progression of their training volumes. The following is some of the generic advice that we give to patients.

- You should not begin to run before your heel has been pain-free for 4 weeks and you can walk 10 km without pain during the walk or the morning after.
- If you need to wear dress shoes, you might consider using soft silicone gel heel pads or contoured prefabricated foot orthoses in your shoes.
- It is important that you decrease or avoid activities that cause your heel pain to worsen.
- When you want to start activities again, you should be careful and progress slowly.

If your patient is a factory worker with many hours of standing work each day, it is important that you discuss potential options on how to decrease the number of hours of standing per day. The following points of

advice are generic and should be individualised to the patient.

- Advise the patient to wear appropriate footwear (offering both cushioning for the heel and support for the rearfoot and midfoot). Although there is little evidence for footwear, it is a good starting point for long-term treatment.^{36, 41}
- Two types of taping have been advocated. The first is low-Dye taping (Fig. 43.5a), which involves the application of rigid tape to the plantar aspect of the foot with the aim of supporting the plantar fascia.^{37, 38} This is sometimes augmented with the second type, calcaneal taping (Fig. 43.5a).
- Recommend appropriately contoured prefabricated foot orthoses.^{39, 40}
- Provide strengthening exercises for proximal muscle groups (e.g. tibialis posterior and hip abductors/external rotators).
- A silicone gel heel pad may be beneficial (Fig. 43.5b) although the evidence is unclear.
- Recommend stretching of the plantar fascia⁴²⁻⁴⁴ (Fig. 43.5c), gastrocnemius and soleus.
- Suggest self-massage with a frozen bottle or golf ball (Fig. 43.5d). This is commonly used but there is no supporting evidence.
- Suggest soft tissue therapy (minimal evidence), both to the plantar fascia (Fig. 43.5e) and proximal myofascial regions including calf, hamstrings and gluteals.⁴⁵

Table 43.3 Evidence table

Interventions	
Conservative	Non-conservative
Likely to be beneficial • Taping (e.g. low-Dye): limited evidence for up to 1 week • Customised and prefabricated foot orthoses (up to 12 months): note that there is no difference between customised and appropriately contoured prefabricated orthoses	Likely to be beneficial • Corticosteroid injection (short-term effects): ultrasound-guided injections have been found to be more effective than placebo
Unknown effectiveness • Heel pads • Night splints • Stretching exercises	Unknown effectiveness • Corticosteroid injection plus local anaesthetic (short term) • Extracorporeal shock wave therapy • Surgery
Likely to be ineffective or harmful	Likely to be ineffective or harmful • Corticosteroid injection (long-term effects) • Corticosteroid injection plus local anaesthetic injection (long-term effects)

Note: No evidence from high-quality systematic reviews is available for dry needling, strengthening exercises, weight loss and non-steroidal anti-inflammatory drugs (NSAIDs) so they have not been included in this table.

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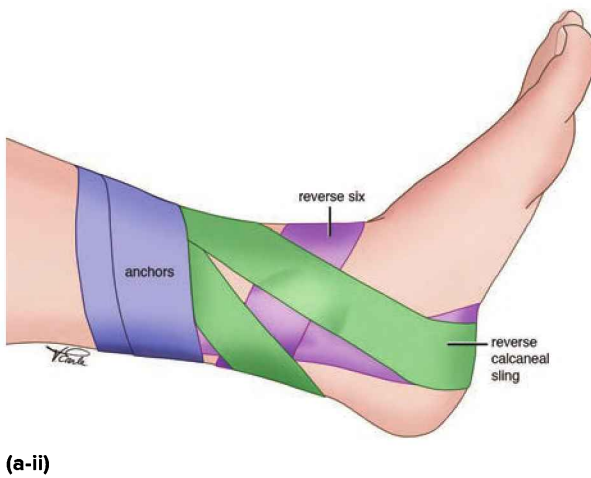
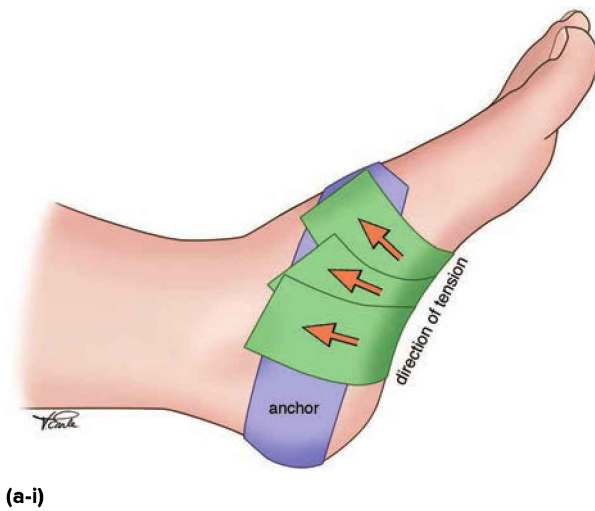


Figure 43.5 Treatment options for plantar fasciopathy (a) Low-Dye taping: (i) standard; (ii) augmented tape applied over standard low-Dye taping (b) Silicone gel heel pad and cup (c) Stretching of the plantar fascia (d) Self-massage with a golf ball (e) Soft tissue therapy to plantar fascia



(f)



(g)

Figure 43.5 (cont.) (f) Corticosteroid injection (g) Strasbourg sock

- Recommend the use of non-steroidal anti-inflammatory drugs (NSAIDs)⁴⁶
- Advise the patient to lose weight. Although there is no evidence for weight loss as being effective, it is biologically plausible that it may be effective given that plantar fasciopathy is highly associated with increased BMI.

Second tier/medium term

The following are recommended second-tier interventions.

- Dry needling of related lower limb muscles is effective.⁴⁷
- Customised foot orthoses may be considered; although there is no evidence that customised foot orthoses are more beneficial than appropriately contoured prefabricated foot orthoses.
- Corticosteroid injection (Fig. 43.5f): evidence supports the use of corticosteroid injection for pain relief in the short term (4–6 weeks).^{35, 48, 49} Ultrasound-guided injections were more effective than palpation-guided injections.⁵⁰ It should always be combined with other treatments such as stretching, foot orthoses or other evidence-based treatments to prevent recurrence. There is some concern that corticosteroid injections are associated with an increased risk of rupture and fat pad atrophy.^{51, 52} However, this is likely related to the solubility and duration of the corticosteroid. Lateral plantar nerve injury is a potential adverse event.⁵³
- High-load strength training consisting of unilateral heel-raises with a towel inserted under the toes to further activate the windlass mechanism may potentially be effective for longstanding cases of plantar fasciopathy. This has been evaluated in one RCT.⁵⁴

Third tier/long term

The following are recommended third-tier interventions.

- Night splints^{55–57} or a Strasbourg sock (Fig. 43.5g) should be considered for patients with pain of over 6 months in duration, although the evidence is unclear and some patients will not tolerate them. One RCT found this to be effective in reducing pain that had been present for more than 6 months.¹
- Extracorporeal shockwave therapy (ESWT) may be of benefit for longstanding cases, although the evidence is still unclear and it may cause substantial pain for some patients while being applied.^{35, 58}
- As the evidence for surgery is unclear and of low quality, it should only be considered in severe longstanding cases that do not respond to any of the evidence-based treatments available. Surgical options include plantar fasciotomy ± nerve release, plantar fasciectomy with neurolysis of the nerve to the abductor digiti quinti muscle,⁵⁹ and minimally invasive endoscopic plantar fascia release.^{60, 61}

Fat pad contusion

The heel fat pad, elastic fibrous tissue septa separated by closely packed fat cells, absorbs the shock of heel strike. Contusion can occur acutely after a fall onto the heels from a height, but is more common chronically because of poor heel cushioning or repetitive stops, starts and changes in direction.

The patient often complains of marked heel pain, particularly during weight-bearing activities. The pain is felt laterally in the heel due to the pattern of heel strike, which may help differentiate this condition from plantar fasciopathy. Examination reveals tenderness in

the posterolateral heel region. MRI often demonstrates oedematous changes in the fat pad, with ill-defined areas of decreased signal intensity on T1-weighted images that increase in signal intensity on T2-weighted images.

The patient should avoid aggravating activities. NSAIDs or paracetamol (acetaminophen) may provide short-term pain relief. Athletes can maintain cardiovascular fitness by cross-training with low-impact activities such as cycling and swimming. Training on soft surfaces can assist athletes with returning to sport.

As the pain settles, the use of a silicone gel heel pad (Fig. 43.5b) and good footwear are important as the athlete resumes activity. A well-fitting heel counter in athletic footwear can reduce shear stress and tension in the fat pad.⁶² Heel-lock taping (see Fig 41.11b) will often provide symptomatic relief.

Calcaneal stress fracture

Calcaneal stress fracture is the second most common tarsal stress fracture. It occurs at either the upper posterior margin of the calcaneum or adjacent to the medial tuberosity, at the point where calcaneal spurs occur. First described among the military as a marching injury (Chapter 45), calcaneal stress fractures occur in runners, ballet dancers and jumping athletes.

Patients give a history of insidious onset of heel pain that is aggravated by weight-bearing activities, especially running. Examination should include viewing the patient perform the causative activity if feasible (i.e. gait analysis in a runner). Palpation reveals localised tenderness over the medial or lateral aspects of the posterior calcaneus and pain that is reproduced by squeezing the posterior aspect of the calcaneus from both sides simultaneously.

Investigations

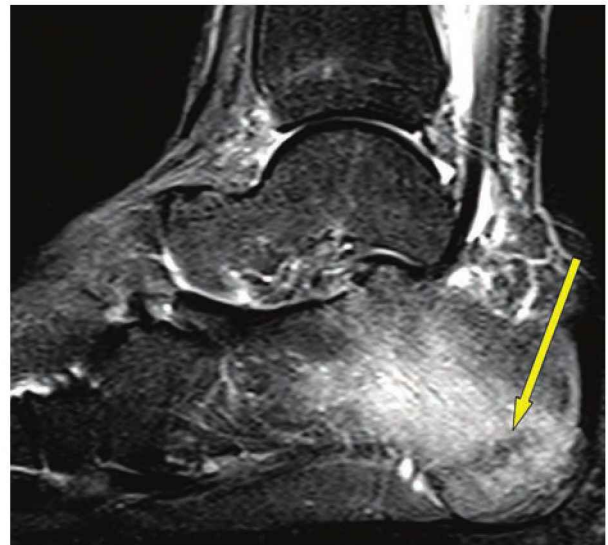
Plain radiography is often normal, but may show a typical sclerotic appearance on the lateral radiograph, parallel to the posterior margin of the calcaneus (Fig. 43.6a). Stress injury was only visible in 15% of plain radiographs.⁶³ The majority of calcaneal stress fractures are posterior. MRI reveals an area of high signal on a T2-weighted image (Fig. 43.6b).

Use the level of pain to guide how much the patient must reduce activity. If pain is marked, a short period of non weight-bearing may be required. Soft heel pads can contribute to immediate pain relief. Weight-bearing can be reintroduced gradually once pain settles.

As with every training-related injury, assess sport technique to determine if abnormal biomechanics may have contributed to the injury. If indicated, prescribe rigid or soft orthoses. Expert foot clinicians consider that joint mobilisation and stretching of the calf muscles and plantar fascia contribute to recovery.



(a)



(b)

Figure 43.6 (a) Plain radiograph appearance of a calcaneal stress fracture. The white arrowheads indicate typical sclerotic appearance (b) T2-weighted MRI showing characteristic appearance³³ of a stress fracture

(b) IMAGING IN SPORTS-SPECIFIC MUSCULOSKELETAL INJURIES 2016, P. 655; ALI GUERMAZI, FRANK W ROEMER, MICHEL D CREMA. © SPRINGER INTERNATIONAL PUBLISHING SWITZERLAND 2016. WITH PERMISSION OF SPRINGER

First branch of lateral plantar nerve (Baxter's nerve) entrapment

An overlooked cause of rearfoot pain is entrapment of the first branch of the lateral plantar nerve (Fig. 43.7). This

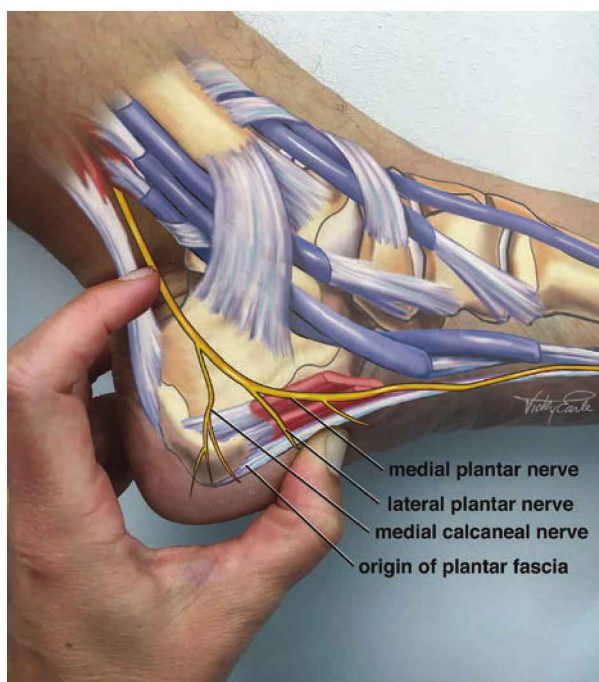


Figure 43.7 The lateral plantar nerve. The nerve is one of three terminal branches of the posterior tibial nerve arising distal to the tarsal tunnel

terminal branch of the posterior tibial nerve arises distal to the tarsal tunnel and it can become trapped between the deep fascia of the abductor hallucis longus and the medial caudal margin of the quadratus plantae muscle.⁶⁴

Pain radiates to the medial inferior aspect of the heel and proximally into the medial ankle region. Patients do not normally complain of numbness in the heel or foot. Often confused with plantar fasciopathy, the two conditions were shown to coexist in 52% of patients with this nerve entrapment.⁶⁵

Examination is nonspecific with maximal tenderness at the plantar medial heel just proximal to the plantar fascia. A Tinel's sign at this location may reproduce pain. Theoretically, the condition is associated with 'subtle weakness of fifth toe abduction'⁶⁶ and atrophy to the abductor digiti quinti muscle, although this is an example of a clinical test that would have very low sensitivity if tested formally (Chapter 14). Some authors claim that atrophy can be identified on MRI or ultrasound exam.^{67, 68}

In the clinic, a diagnostic injection with local anaesthetic, ideally performed using ultrasound to identify the nerve, can increase the clinician's confidence in the diagnosis (increased positive likelihood ratio, Chapter 14).⁶⁹ Referral for nerve conduction tests may be helpful for diagnosis and to determine the location of compression in the lateral plantar tunnel, Baxter's nerve or more proximally in the tarsal tunnel.⁶⁷ This is particularly important if surgery is being considered.

Treatment consists of using soft-soled shoes, heel cups or arch support, tape or an orthosis for athletes with excessive pronation. A corticosteroid injection, best ultrasound-guided as described earlier, may be helpful. Radiofrequency denervation of the nerve may give short-to medium-term pain relief. Surgical release would be indicated if conservative measures have failed.⁶⁴

MIDFOOT PAIN

The midfoot consists of the three cuneiform bones, the cuboid and the navicular bones as well as the surrounding soft tissues (Figs 43.1 and 43.2). The most common cause of midfoot pain is midtarsal joint sprain after ankle injury, but the most important cause of midfoot pain is stress fracture of the navicular bone. Tendinopathy of the extensor tendons is another common cause of midfoot pain. The causes of midfoot pain are listed in Table 43.4.

Table 43.4 Causes of midfoot pain in order of importance (clinical relevance)

Common	Less common	Not to be missed
Navicular stress fracture	Cuneiform stress fracture	Lisfranc's joint injury (fracture or dislocation)
Midtarsal joint sprain	Cuboid stress fracture	Osteoid osteoma
Lisfranc joint injury (sprain)	Stress fracture base second metatarsal	Complex regional pain syndrome type 1 (after knee or ankle injury)
Tibialis posterior tendinopathy	Extensor tendinopathy	Referred pain
Plantar fascia strain	Peroneal tendinopathy	
	Abductor hallucis strain	
	Cuboid syndrome	
	Tarsal coalition (in adolescents)	
	Köhler's disease (in young children)	
	Accessory navicular bone	

Clinical approach to midfoot pain

Acute midfoot pain results from a sprain of the midtarsal joint or plantar fascia. Gradual pain is a sign of overuse injury, such as extensor tendinopathy, tibialis posterior tendinopathy or navicular stress fracture. In most conditions, pain is well localised to the site of the injury, but in navicular stress fracture, pain is poorly localised.

Examination involves palpation of the area of tenderness and a biomechanical examination to detect factors that predispose to injury. Examination of the midfoot is shown in Figure 43.8.



(a)



(b)

Figure 43.8 Examination of the midfoot (a) Palpation—N spot. The proximal dorsal surface of the navicular is tender when a stress fracture is present (b) Palpation—extensor tendons. These may be tender as they pass under the extensor retinaculum

Investigations

If there is a clinical suspicion of a stress fracture of the navicular or a cuneiform, plain radiographs are indicated. This rarely reveals a fracture, even if one is present, but it is useful to rule out tarsal coalition, to show bony abnormalities such as talar beaking (osteophytes at the talonavicular joint) and accessory ossicles, and to exclude bony tumours. MRI should be performed if X-ray fails to reveal a stress fracture.

Navicular stress fracture

Stress fractures of the navicular are among the most common stress fractures seen in the athlete, especially in sports that involve sprinting, jumping or hurdling.⁷⁰ The stress fracture commonly occurs in the middle third of the navicular bone (Fig. 43.9). This site is a relatively avascular 'watershed area' of bone as the blood supply comes from the lateral and medial sides of the navicular. This is thought to contribute to poor healing of these stress fractures.

Spikes in training load are thought to be the most common cause of navicular stress fractures. It is likely that the navicular bone is jammed between the talus proximally and the cuneiforms distally. Many clinicians speculate that reduced ankle dorsiflexion may cause excessive midfoot dorsiflexion and thus increase the load at the talonavicular joint with subsequent failure of the slightly concave proximal lip of the navicular at that site.⁷¹

Clinical features

The onset of symptoms is usually insidious, consisting of a poorly localised midfoot ache associated with activity. The pain typically radiates along the medial aspect of the

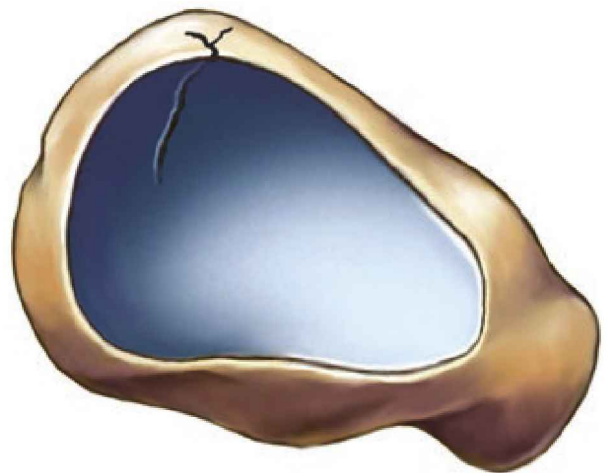


Figure 43.9 Navicular stress fractures occur at a characteristic proximal site on the bone. This N-spot corresponds with the point of tenderness on palpation

longitudinal arch or the dorsum of the foot. The symptoms abate rapidly with rest, unfortunately often contributing to diagnostic delays. Examination reveals localised tenderness at the N-spot: the proximal dorsal portion of the navicular.⁷² If palpation confirms tenderness over the N-spot, particularly when compared to a symmetric location on the contralateral foot (Fig. 43.8a), the athlete should be considered to have a navicular stress fracture until proven otherwise.

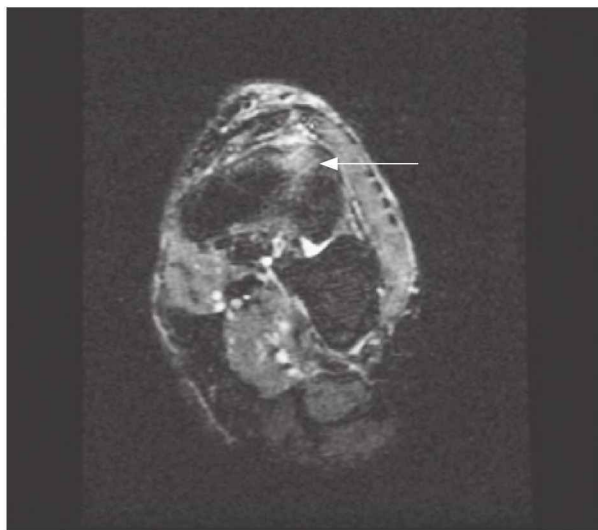
PRACTICE PEARL

The pain of navicular stress fracture is often poorly localised.

The astute clinician assesses the patient for factors that predispose to the condition, such as tarsal coalition, excessive pronation and restricted ankle dorsiflexion. Any factors that need to be corrected should be attended to at the appropriate time in the management program (see below).

Investigations

Plain radiographs lack sensitivity to detect navicular stress fractures.⁷² MRI (Fig. 43.10a) is the investigation of choice. Computed tomography (CT) scan can further clarify the morphology of this fracture (Fig. 43.10b and c). Poor positioning of the patient's foot and scanning with slice thickness >2 mm can lead to a navicular stress fracture being missed on CT scan.⁷³



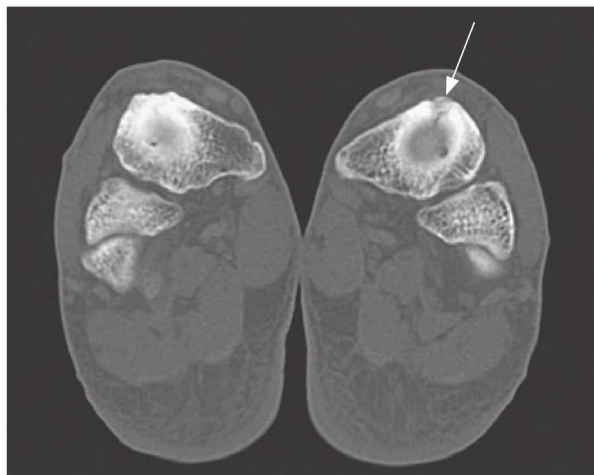
(a)

Management of navicular bone stress injuries

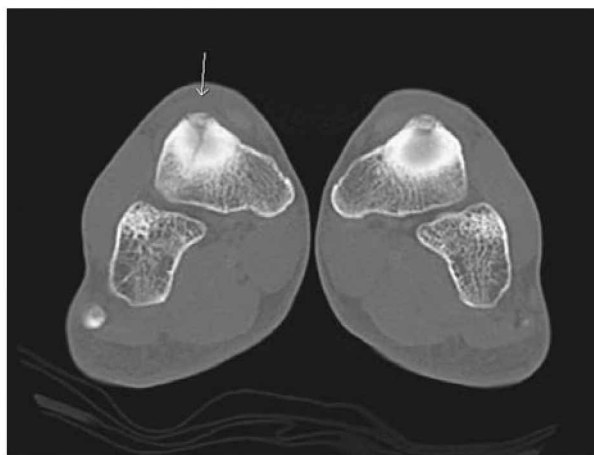
We consider treatment according to whether the diagnosis is: 1. stress reaction; 2. typical (uncomplicated) stress fracture; and 3. complicated stress fracture (e.g. delayed diagnosis, delayed union). (See also Chapter 4–Bone stress injuries.)

Stress reaction

If a patient has only had symptoms for a relatively short time (days, <2 weeks), he or she may not have a full



(b)



(c)

Figure 43.10 Navicular stress fracture (a) T2-weighted MRI showing bone marrow oedema (b) CT scan showing undisplaced fracture (arrow) (c) CT scan showing more extensive Y-shaped fracture

stress fracture but only a stress reaction. A positive MRI combined with a normal CT scan (skilfully executed, as mentioned in the previous section) confirms the diagnosis of navicular stress *reaction* (no cortical breach). This can be treated with weight-bearing relative rest which means the patient can take weight through the part but cannot play sport. This is generally done by having the patient wear a CAM boot for 1–3 weeks until pain is absent and the N-spot is no longer tender. This is then followed by a graduated return to activity (aim for increases in the acute:chronic workload to be <1.1 or 10%) over 2–3 months (Chapter 12).

Typical stress fracture

In the majority of cases, patients who present with midfoot pain and navicular tenderness have a frank stress fracture visible on both MRI and CT scan. Remember that bone pathology (failure) begins before the patient feels pain, so day 1 of pain is not day 1 of relative excessive load and microfractures (which precede frank stress fracture).

PRACTICE PEARL

The typical non-displaced stress fracture should be treated with absolute non weight-bearing.

Non weight-bearing can be achieved with patient education and a rigid fibreglass boot.⁷² This period of non weight-bearing is ideally carried through a *minimum* of 6 weeks but many clinicians recommend 8 weeks so as to lower the risk of recurrence.

At the end of the period of non weight-bearing immobilisation, the boot should be removed and the N-spot palpated for tenderness. Generally, the N-spot will be non-tender; however, if tenderness is present, the patient should have a further 2 weeks of non weight-bearing immobilisation. Management is primarily based on the clinical assessment, as there is poor CT and MRI correlation with clinical union of the stress fracture (Fig. 43.11).^{70, 74}

Following removal of the boot, it is essential for a skilled clinician to mobilise the stiff ankle and the subtalar and midtarsal joints. The calf muscles require soft tissue therapy and a progressive strengthening program. This rehabilitation must precede return to running.

Activity must begin gradually, slowly building up to full training over a period of 6 weeks. Return to sport often takes 3 months in the ideal case where the athlete has maintained fitness with cross-training and meets all his or her rehabilitation landmarks smoothly. Taking 4 months to return to play is within normal limits for this injury.⁷⁵ Note that even if a patient rested for 8 weeks



Figure 43.11 Imaging of navicular stress fracture does not necessarily mirror clinical recovery. This CT scan of a 26-year-old runner shows that the fracture line is still evident (arrow) but the patient had returned pain-free to all competition and had no further recurrences of stress fracture

(without any injury), it would take 3 months to return to a high level of training merely to avoid spiking training loads.⁷⁶

PRACTICE PEARL

WARNING: A recipe for navicular disaster is for the patient to think that 'non weight-bearing' is merely an aspirational goal or a guideline. Countless athletes have had their careers ruined by navicular stress fracture recurrence. The fracture is notorious for non-union if the first treatment attempt is a failure.⁷²

Complicated stress fracture

Often, patients with a stress fracture of the navicular present after a long period of pain or after a period of weight-bearing rest. There is no evidence to suggest these patient should have immediate surgical treatment. Patients who have been unsuccessfully treated with weight-bearing rest or short-term boot immobilisation should undergo non weight-bearing immobilisation for 8 weeks. This method of

treatment is associated with an 80% successful return-to-sport rate and may be successful even in cases where the patient has had longstanding (>6 months) pain.^{70, 74}

Failure of rigorous conservative treatment is an indication for surgery. Which type of surgery is best has not been tested. Some surgeons favour internal fixation, others prefer to excise the fracture and perform a bone graft. As the fracture involves a joint surface (proximal navicular, at the talonavicular joint; Fig. 43.10) surgery is not universally successful.

A 2010 systematic review and meta-analysis found no significant difference between non weight-bearing conservative treatment and surgical treatment in terms of pain reduction.^{74, 77}

Midtarsal joint sprains

Themidtarsal joint (Chopart's joint) consists of the talonavicular and calcaneocuboid joints. Other joints in themidtarsal area are the naviculocuneiform, cuboid cuneiform and intercuneiform joints. Injuries to themidtarsal joints are most commonly seen in gymnasts, jumpers and footballers.

Individual ligamentous sprains to themidtarsal joints are uncommon and if they occur usually affect the dorsal calcaneocuboid or bifurcate ligament, comprising the calcaneonavicular and calcaneocuboid ligament.

Dorsal calcaneocuboid ligament injury

Patient presents with pain in the lateral midfoot following an inversion injury. Examination reveals localised tenderness and swelling at the dorsolateral aspect of the calcaneocuboid joint. Stress inversion of the foot elicits pain.

Plain radiography is required to exclude fracture and MRI may confirm the diagnosis. Taping may provide additional support while orthoses may help to de-load the painful ligaments. A joint sprain can sometimes lead to secondary joint inflammation. This generally responds well to NSAIDs but if inflammation persists the patient may benefit from a corticosteroid injection into the painfulmidtarsal joints.

Bifurcate ligament injuries

Injuries to the bifurcate ligament may be associated with fractures to the anterior process of the calcaneus secondary to violent dorsiflexion, forceful plantarflexion and inversion injuries (Chapter 41). Patients present with lateral midfoot pain and swelling, usually following an ankle sprain or injury. Examination reveals local tenderness and occasionally swelling overlying the ligament, with pain elicited at the site with simultaneous forefoot supination and plantarflexion.

Plain radiograph may reveal a fracture of the anterior process of the calcaneus but it can miss one, so never rely on X-ray to rule out a fracture. An MRI scan can be used to confirm the joint/ligament sprain. If a fracture is present, a CT scan may be required for further assessment (see Fig. 41.17).

Treatment is similar to the dorsal calcaneocuboid sprain mentioned previously. If a non-displaced or mildly displaced fracture of the anterior process of the calcaneum is present, 4 weeks immobilisation is required. If the fracture is displaced, surgery is required.

Lisfranc joint injuries

The eponymous Lisfranc joint refers to the tarsometatarsal joints: the bases of the five metatarsals with their corresponding three cuneiforms and cuboid (Fig. 43.12). The spectrum of injuries of the Lisfranc joint complex ranges from partial sprains with no displacement, to complete tears with separation (diastasis) of the first and second metatarsal bones and, depending on the severity, different patterns of tarsal and metatarsal displacement (Fig. 43.13).

PRACTICE PEARL

Be alert! Although Lisfranc joint injuries ("midfoot sprains") are not common in the general population, they are the second most common foot injury in athletes. In sport, they generally occur via low-velocity indirect force.

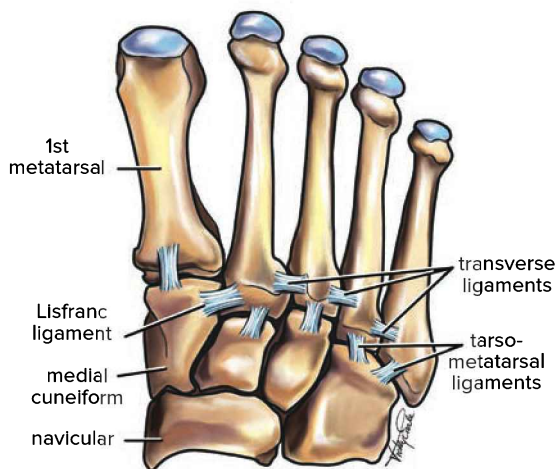


Figure 43.12 Ligamentous attachments of the Lisfranc joint articulation

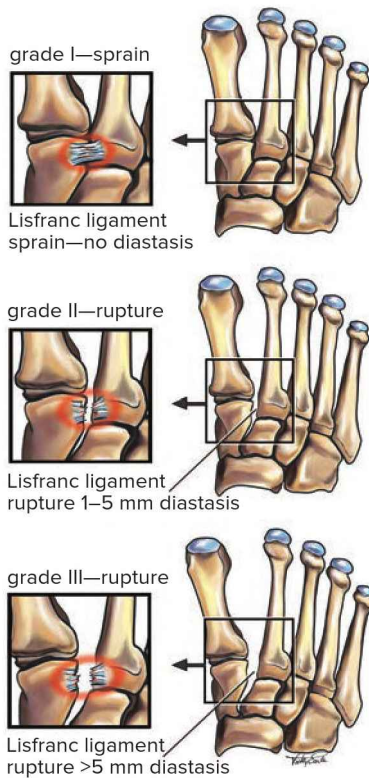


Figure 43.13 Lisfranc joint injury classification system

Lisfranc joint fracture-dislocation is rare in sport, but, because of its disastrous consequences if untreated, the diagnosis must be considered in all cases of 'midfoot sprain' in the athlete.

Causes

There are two main mechanisms of injury.

- **Direct.** This injury is relatively uncommon and occurs as a simple crush injury to the tarsometatarsal joint region. There is no specific pattern of damage or distinctive appearance with a direct injury.
- **Indirect.** This mechanism is more common and generally occurs secondary to a longitudinal force sustained while the foot is plantarflexed and slightly rotated. There are three common injury situations: longitudinal compression (Fig. 43.14); a backward fall with the foot entrapped; and a fall on the point of the toes.⁷⁸

The extent of the damage depends on the severity of the injury. In milder injuries, the weak dorsal tarso-metatarsal ligaments are ruptured; with more severe injuries, there may also be fractures of the plantar aspect of the metatarsal base, or the plantar capsule may rupture and the metatarsal may displace dorsally. Thus, a fracture

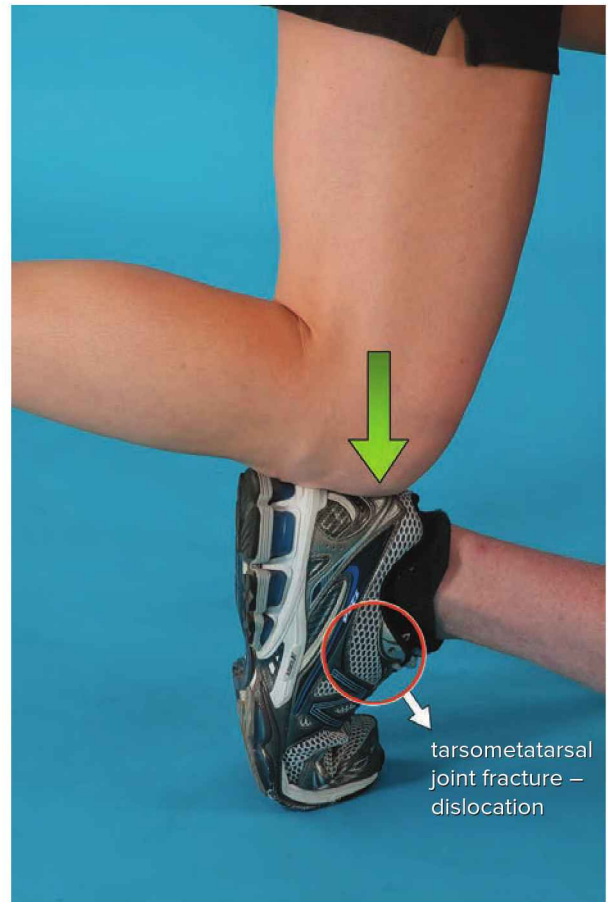


Figure 43.14 Mechanism of injury to Lisfranc joint—longitudinal compression

at the plantar base of a metatarsal can be a clue to a subtle Lisfranc joint injury.

Clinical features

A patient with Lisfranc joint injury may complain of midfoot pain and difficulty weight-bearing, following an acute injury by the mechanisms described above. Pain is classically aggravated by forefoot weight-bearing—the patient is unable to run on the toes and feels pain on the push-off phase of running (and sometimes during walking) and on calf raises. Often the presentation may be delayed, and the patient presents with ongoing midfoot pain and swelling, aggravated by running.

PRACTICE PEARL

Midfoot pain that persists for more than 5 days post injury should raise suspicion of a Lisfranc joint injury.

Examination reveals:

- tenderness with or without swelling on the dorsal midfoot, often with associated bruising in this region
- pain with combined eversion and abduction of the forefoot while the calcaneus is held still
- plantar medial ecchymosis may be observed in a delayed diagnosis⁷⁹
- neurovascular examination is mandatory as the dorsalis pedis artery can be compromised in the initial injury or by subsequent swelling of the foot.⁷⁸

Investigations

Plain weight-bearing radiographs are recommended. Diastasis between the first and second metatarsal bases of greater than 2 mm. (Fig. 43.15a) suggests a Lisfranc joint injury, although in patients with a metatarsus adductus a 3 mm separation may be normal. In such cases, it is essential to take comparative weight-bearing X-rays of the non-injured side, as a difference in diastasis of greater than 1 mm between the two sides is considered diagnostic.

Other radiological signs that may indicate an injury to the Lisfranc joint include a 'fleck sign', which appears as a fleck fracture near the base of the second metatarsal or medial cuneiform or, in the lateral view, either dorsal displacement of the metatarsal bases relative to the tarsus or flattening of the medial longitudinal arch. However, a dislocation may reduce spontaneously, and the foot may appear normal on plain X-rays despite severe soft tissue disruption. There was a high correlation between rupture or grade II sprain of the plantar ligament between the first cuneiform and the bases of the second and third metatarsals, and true midfoot instability.

If injury is suspected, and X-ray is normal, both CT (Fig 43.15b) and MRI (Fig 43.15c) can help the clinician make the diagnosis. MRI is more sensitive and specific than CT to demonstrate all the components of the Lisfranc complex.^{80, 81}

Static and dynamic ultrasound examination of the dorsal cuneiform metatarsal (C1-M2) ligament is a sensitive indicator of stable or unstable sprains. Thickening to this ligament without joint space widening is consistent with a grade 1 sprain. Absence of this ligament and joint space wider than 2.5 mm, plus increased spacing upon weight-bearing, suggests Lisfranc ligament tear.⁸²

Treatment

The treatment of a Lisfranc joint injury depends on the degree of instability present.⁸³ In grade I injuries, where there is no instability (diastasis), conservative management is recommended with non weight-bearing in a CAM boot or Aircast® for 6 weeks. The goal of this treatment is to restore the integrity of the tarsometatarsal ligaments and



(a)

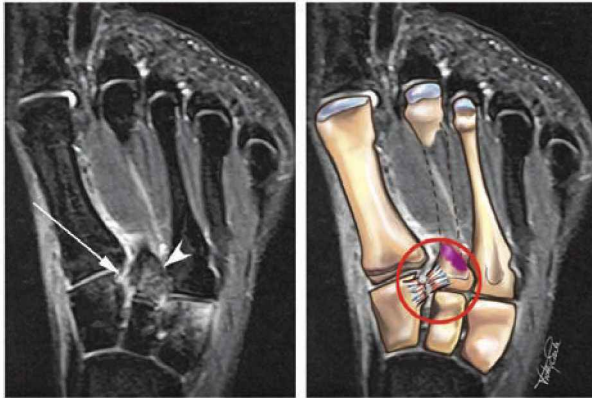
Figure 43.15 Lisfranc fracture–dislocation (a) Plain radiograph

hence the stability of the midfoot. Following removal of the cast, mobilisation of the ankle and a calf-strengthening program are required. Orthoses may be needed to correct the intrinsic alignment of the foot and to support the second metatarsal base. A graded return to activity is required. Recovery time from diagnosis to activity return is prolonged, often requiring 3–4 months of total treatment.⁷⁹

Due to instability, injuries of grade II or greater severity require surgical reduction and fixation. This may be performed percutaneously or, in more difficult cases, an open operation may be needed. There has also been consideration for primary arthrodesis as opposed to open reduction internal fixation (ORIF); however, this is a less-preferred method.⁸⁴ In the situation of a delay in diagnosis, similar treatment protocols are required, however grade II or grade III Lisfranc injury is a significant injury that has a much better prognosis if managed correctly initially, rather than being salvaged once there is prolonged joint malalignment and non-union.



(b)



(c)

Figure 43.15 (cont.) (b) CT scan showing fractures of middle and lateral cuneiforms (arrows) with 0.30 cm of diastasis in the Lisfranc joint (c) Axial STIR MRI highlights torn Lisfranc ligament (arrow) and associated cuboid and metatarsal base contusion (arrowhead)³³

(c) IMAGING IN SPORTS-SPECIFIC MUSCULOSKELETAL INJURIES, 2016, P. 660: ALI GUERMAZI, FRANK W ROEMER, MICHEL D CREMA. © SPRINGER INTERNATIONAL PUBLISHING SWITZERLAND 2016. WITH PERMISSION OF SPRINGER

A delay in diagnosis has been associated with a poor outcome, a prolonged absence from sport, and chronic disability due to ligamentous instability of the tarsometatarsal joint.

Tibialis posterior tendinopathy (distal pain presentation)

Tibialis posterior tendinopathy is described in the previous chapter on ankle pain (Chapter 42), but tendinopathy at the distal insertion of the tendon into the navicular may present primarily as medial foot pain. In younger patients, the accessory ossicle, the os navicularis (Fig. 43.16), may be avulsed. This requires orthoses and gradual return to activity. Surgery is rarely indicated as the tibialis posterior has a complex insertion into all bones of the foot.

Less common causes of midfoot pain

First tarsometatarsal joint pain

A prominent first metatarsal-cuneiform joint may cause pain to the dorsomedial midfoot. Most often, constricting shoelaces or excessively tightly fitting shoes will compress a dorsally protuberant first tarsometatarsal joint. The prominent base of the first metatarsal may be anatomically determined or developed due to degenerative changes at the joint. Dorsal cutaneous nerve compression can also result from this prominent dorsal bump. Suggested treatments focus to reduce pressure at this site. Modified lacing patterns, shoes with soft leather through the midfoot,



Figure 43.16 The os navicularis (arrow)

and accommodative padding around the bump can all be attempted before more aggressive treatments. Surgery may be required to remove persistently painful bony exostoses.

Extensor tendinopathy

The extensor (dorsiflexor) muscles of the foot comprise the tibialis anterior, extensor hallucis longus and brevis, as well as extensor digitorum longus and brevis. Tibialis anterior tendinopathy is the most common tendinopathy occurring in the extensor muscles of the foot. Tendinopathies of the extensor hallucis longus and brevis and extensor digitorum longus and brevis muscles are rare.

The tibialis anterior muscle eccentrically controls plantarflexion and eversion of the foot at heel strike and its tendon is therefore susceptible to overuse injury. Tendinopathy generally occurs because of excessive training load in relation to background preparation. Experienced clinicians believe that stiffness of the first MTP joint and midfoot joints may contribute.

Generally, after a period of overuse, the patient with extensor tendinopathy complains of burning or aching medial midfoot pain. Examination may reveal tenderness, often with mild swelling, at the insertion of the tibialis anterior tendon at the base of the first metatarsal and the cuneiform. Pain may be elicited by passive plantar flexion combined with eversion, abduction and dorsiflexion. This is otherwise known as the tibialis anterior passive stretch (TAPS) test.⁸⁵ Resisted dorsiflexion and eccentric inversion may also exacerbate symptoms. Functional assessment may reveal excessive heel strike or over-striding during running. Both ultrasound and MRI may reveal swelling of these tendons and exclude the presence of a degenerative tear.

Popular clinical treatment includes relative rest and soft tissue therapy to the extensor muscles. Extensor muscle strengthening is advocated as is the case with other tendinopathies. The underlying precipitating cause needs to be addressed. If this is load, then training load needs to be considered. Local treatment may include mobilisation of the first ray, and tarsometatarsal and midtarsal joints, if the first MTP joint and midfoot is stiff. Modification of running technique may be required (reduce stride length) and footwear may need to be replaced.

Peroneal tendinopathy

Peroneal tendinopathy is an overuse injury of the peroneal tendons associated with running, dancing and in those with chronic lateral ankle instability. The management of this condition is described in Chapter 42.

Abductor hallucis strains

Abductor hallucis strains result in pain along the medial longitudinal arch and are often associated with excessive pronation. The abductor hallucis tendon is tender to

palpate. Treatment consists of local electrotherapeutic modalities and taping.

Cuboid syndrome

The renowned physiotherapist from the American Ballet Theatre in the 1990s, Peter Marshall, brought this condition to the mind of clinicians who specialise in foot pain.^{86, 87} Marshall argued that due to excessive traction of the peroneus longus, the cuboid becomes 'subluxated'. The term 'subluxated' is still challenging, 35 years after Marshall used it; certainly there have never been imaging changes captured to confirm subluxation or dislocation. Cuboid syndrome may also manifest following an acute plantarflexion and inversion ankle injury, and is common among professional ballet dancers. Pain is experienced with lateral weight-bearing. Most patients with this syndrome have excessively pronated feet but it is also seen in patients with lateral ankle instability.

The peroneus longus subluxates the cuboid bone so that the lateral aspect is rotated dorsally and medially. There may be a visible depression over the dorsal aspect of the cuboid. Treatment involves a single manipulation to reverse the subluxation.⁸⁸ The cuboid should be pushed upwards and laterally from the medial plantar aspect of the cuboid, as shown in Figure 43.17. In the largest case series, athletes returned to sport immediately following cuboid manipulation without any recurrences.⁸⁹

Stress fracture of the cuboid

Stress fractures of the cuboid are uncommon and occur secondary to compression of the cuboid between the calcaneus and the fourth and fifth metatarsal bones when exaggerated plantarflexion is undertaken with or without inversion. Diagnosis is often delayed due to non-specific signs or symptoms clinically and typically normal plain radiography. MRI is the preferred choice of imaging (Fig. 43.18).^{90, 91} Treatment of the uncomplicated stress fracture requires non-weight-bearing for 4–6 weeks, followed by a graduated return to activity.

Stress fracture of the cuneiforms

Stress fractures of the cuneiform bones are quite rare. We reported one in 1993 and have seen less than a handful between us since then.⁹² Theoretical risk factors are large body habitus, plantar fascia injury,⁹³ overuse and altered gait pattern. Limited weight-bearing rest is treatment for medial cuneiform stress fractures; however, stress fractures of the intermediate cuneiform require surgical reduction and fixation for adequate healing.⁹⁴

Köhler's disease

Köhler's disease, or osteochondritis of the navicular, is found in children aged 2–8 years (Chapter 44).



Figure 43.17 Manipulation of the cuboid in a superior and lateral direction in cuboid syndrome

Tarsal coalition

Congenital fusions of the foot bones usually present as midfoot pain. Tarsal coalitions may be osseous, cartilaginous (synchondrosis), fibrous (syndesmosis) or a combination. The estimated prevalence is 1–2%, although a recent CT and cadaver dissection study found 12.7%.⁹⁵ A fibrous coalition is relatively mobile and therefore may not cause any pain on limited motion. As fibrous or cartilaginous coalitions ossify during adolescence, rearfoot or midfoot joint range of motion decreases, placing additional stress on the talocrural (ankle) joint. Accounting for 90% of coalitions, the two most common tarsal coalitions occur at the calcaneonavicular joint and the talocalcaneal joint; 50–60% occur bilaterally.

Clinical presentation usually occurs during the adolescent/young adult years coinciding with ossification of the cartilaginous coalitions and more vigorous sports and physical activity.⁹⁶ Patients can also present after minor trauma such as an ankle sprain that does not clinically improve (Chapter 41). Because tarsal coalition causes subtalar restriction, adjacent joints may function abnormally and thus the patient presents with



Figure 43.18 MRI demonstrating stress fracture cuboid (arrow)³³

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joint spurs and symptoms remote from the coalition (Fig. 43.19).

Examination usually reveals limited range of subtalar and midtarsal joint movement that may be painful at end range. Plain radiographs remain the initial test of choice and are the most useful diagnostic tool for osseous coalitions. A Harris-Beath view may detect a talocalcaneal coalition. A complete C-sign seen on a lateral X-ray is highly correlated with bony talocalcaneal coalition (Fig. 43.19), although the absence of this sign does not rule out the possibility of this condition.⁹⁷ CT scan provides the most complete diagnostic imaging for bony tarsal coalition, while MRI can identify fibrous coalition.

Asymptomatic cases do not require treatment. For symptomatic cases, initial conservative management includes a period of relative rest to allow inflamed joints to settle. NSAIDs may provide a short-term palliative effect as can paracetamol (acetaminophen). Orthoses may alter biomechanics favourably but they can also eliminate



Figure 43.19 Radiograph showing talocalcaneal coalition (black arrows) with a continuous cortical line, the C-sign, joining the talus and sustentaculum tali, and talar beaking (white arrow) that results from abnormal subtalar movement

movement at joints that provide motion so they need to be considered carefully and their effects monitored. Corticosteroids are mentioned in sports medicine texts but we do not see a compelling case in tarsal coalition.⁹⁶

Surgery may be required for recalcitrant or disabling pain. As always, an experienced subspecialist foot surgeon is likely to provide better advice and have superior clinical outcomes compared to a general orthopaedic surgeon who is essentially unfamiliar with the intricacies of the procedure or its postoperative rehabilitation.

FOREFOOT PAIN

Forefoot problems that can severely compromise dancers' and athletes' performance include stress fractures and soft tissue injuries such as the plantar plate tear. Common problems that are far more significant in a competitive athlete's life than in a 'weekend warrior' include corns, calluses and nail problems. Causes of forefoot pain are listed in Table 43.5 and that list guides the order in which we discuss the conditions in the chapter.

Clinical perspective

Accurate diagnosis in the forefoot often requires the clinician to combine the evidence from a careful history (mechanisms, time course), with careful palpation and understanding of the limitations of investigations (e.g. Morton's neuroma will likely come with a normal MRI).

History

Most causes of forefoot pain result from training errors and thus have an insidious onset. Occasionally, acute forefoot pain may result from a sprain of the first MTP joint ('turf toe'). Sensory symptoms may indicate that a neuroma is present.

Examination

The key to examination of the patient with forefoot pain is precise palpation to identify the site of maximal tenderness. Biomechanical examination is necessary.

Table 43.5 Causes of forefoot pain

Common	Less common	Not to be missed
Stress fracture of the neck of metatarsals I–IV Stress fracture of the base of the second metatarsal Fractures of the fifth metatarsal MTP joint synovitis First MTP joint sprain ('turf toe') Hallux limitus Hallux valgus ('bunion') Sesamoid injuries including stress fracture Plantar plate tear Morton's neuroma Corns, calluses Plantar wart Onychocryptosis (ingrown toenail)	Stress fracture of the great toe Joplin's neuritis Freiberg's osteochondritis Toe clawing Subungual haematoma Subungual exostosis	Referred pain

1. Observation (Fig. 43.20a)
2. Palpation:
 - (a) metatarsals (Fig. 43.20b)
 - (b) first MTP joint (Fig. 43.20c)
 - (c) sesamoid bone of the foot (Fig. 43.20d)
 - (d) space between third and fourth metatarsal (Fig. 43.20e)

Investigations

Plain radiographs may show evidence of a healing stress fracture or acute fracture, the presence of hallux abducto-valgus, hallux limitus or a subungual exostosis. MRI may confirm the diagnosis of a stress fracture. Both MRI and ultrasound have limited utility to identify a neuroma⁹⁸ (see section on Morton's neuroma below).

Stress fractures of the neck of metatarsals I–IV

Stress fractures of the metatarsals are the second most common stress fractures of all.⁹⁹ Training errors (sudden

increase in load, change in training conditions or footwear) or poor biomechanics can load the forefoot to failure.¹⁰⁰ Fatigue of the gastrocnemius muscle increases forefoot loading.^{101, 102}



(b)



(c)



(d)



(a)

Figure 43.20 Examination of the patient with forefoot pain (a) Observation for the presence of obvious abnormalities (e.g. hallux abducto-valgus, claw toes, Morton's foot, plantar warts, onychocryptosis, corns, callus) (b) Palpation—metatarsals (c) Palpation—first MTP joint (d) Palpation—sesamoid



(e)

Figure 43.20 (cont.) (e) Palpation—space between third and fourth metatarsal

The most common metatarsal stress fracture is at the neck of the second metatarsal. This occurs in the pronating foot, when the first ray is dorsiflexed and the second metatarsal becomes subjected to greater load. The second metatarsal is also susceptible to stress fracture in the case of a Morton's foot, where the second ray is the longest ray (Fig. 43.21). The base of the second metatarsal is firmly fixed in position next to the cuneiform bones, further increasing the likelihood of fracture. Stress fracture of the second metatarsal is common in ballet dancers. Stress fractures of the other metatarsals also occur, particularly in the third metatarsal if it is longer than the second.

Plain radiograph is not sensitive to detect early stress fractures but may reveal a radiolucent line or periosteal thickening if the fracture has been present for a few weeks (Fig. 43.22). MRI is the investigation of choice.

The management of most stress fractures is straightforward, involving rest from weight-bearing aggravating activities for approximately 4 weeks, and addressing any

underlying contributing factors. If the patient is required to be on his or her feet excessively, the use of a CAM boot may be required for 1-2 weeks until pain settles.

The athlete should be allowed to recommence activity when he or she does not experience pain when walking and there is no local tenderness at the fracture site. A graduated exercise program should be instituted to return the athlete to full training and competition. Orthoses may



Figure 43.21 Morton's foot with the first ray shorter than the second. The patient with a metatarsal stress fracture complains of forefoot pain aggravated by activity such as running or dancing. The pain is not severe initially but gradually worsens with activity. Examination reveals the presence of focal tenderness overlying the metatarsal



Figure 43.22 Stress fracture of the second metatarsal. X-ray appearance—arrow points to new bone formation at the site of the stress fracture. This ballet dancer's metatarsal displays generalised cortical hypertrophy

be required to control abnormal foot mechanics; however, it is still unclear from current research whether orthoses can prevent metatarsal stress fractures.¹⁰³ Any instability during forefoot weight-bearing may predispose to the development of a stress fracture. Intrinsic and extrinsic foot muscle strengthening exercises may help prevent recurrence.

PRACTICE PEARL

Two metatarsal stress fractures require special treatment: fractures of the base of the second metatarsal and the stress fracture in the Jones position of the fifth metatarsal.

Special metatarsal stress fracture: base of the second MT

An unusual fracture of the base of the second metatarsal (Fig. 43.23a) involving the joint occurs in ballet dancers.^{99, 104–106} The fracture can sometimes be seen on plain X-ray (Fig. 43.23b), however MRI is the investigation of choice (Fig. 43.23c). This fracture should be treated by having the dancer remain non weight-bearing on crutches until tenderness settles, usually at least 4 weeks. The differential diagnosis of proximal second metatarsal stress fracture is chronic joint synovitis.

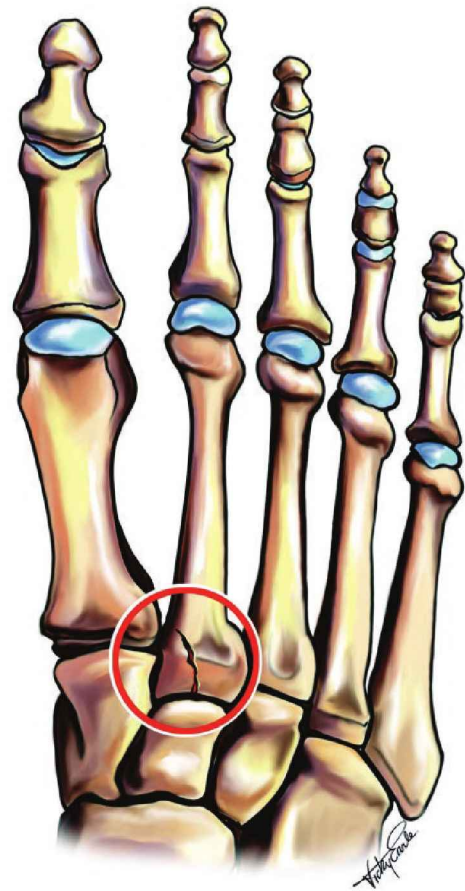
These stress fractures may be associated with Achilles tightness, low bone mass and high training loads.¹⁰⁷

Clinicians expert in managing metatarsal fractures advocate soft tissue therapy to the calf muscles in addition to strengthening of all foot muscles including the peroneals.

Fractures of the fifth metatarsal

Ever since Sir Robert Jones described¹⁰⁸ in 1902 a fracture 2 cm from the base of the fifth metatarsal that he himself sustained (while maypole dancing!), there has been considerable confusion regarding the different types of fractures and their definitions. For many years, fractures at the base of the fifth metatarsal were simply divided into avulsion fractures of the tuberosity, and all others, which were collectively known as Jones fractures.

In 1993, Lawrence and Botte were the first to introduce a classification distinguishing three different fracture types based on mechanism of injury, fracture location, treatment options and outcome.¹⁰⁹ They described the 'Jones fracture' being located more proximally, and assigned it at the level of the entire fourth-fifth intermetatarsal articulation

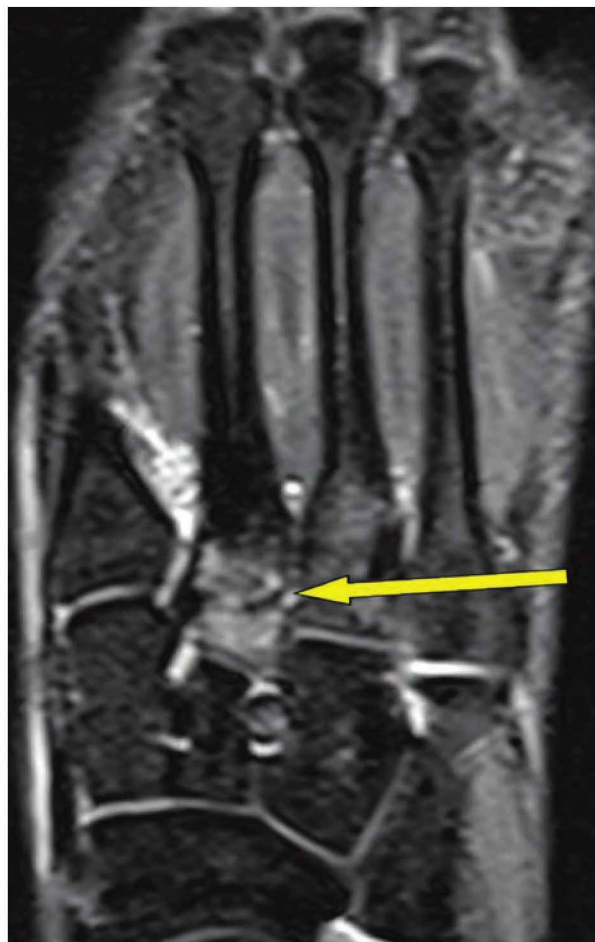


(a)

Figure 43.23 (a) Stress fracture of the base of the second metatarsal



(b)



(c)

Figure 43.23 (cont.) (b) Stress fracture of the base of the second metatarsal involving the joint. The fracture line (arrowhead) extends to the second tarsometatarsal joint (c) MRI of stress fracture (arrow) at base of second metatarsal. This is a common site of stress fractures in ballet dancers³³

(c) IMAGING IN SPORTS-SPECIFIC MUSCULOSKELETAL INJURIES 2016, P. 659, ALI GUERMAZI, FRANK W ROEMER, MICHEL D CREMA. © SPRINGER INTERNATIONAL PUBLISHING SWITZERLAND 2016. WITH PERMISSION OF SPRINGER

(Zone 2). The location formerly known as 'Jones fracture' at the proximal diaphysis was renamed as 'diaphyseal stress fracture' (Zone 3). Fractures to the most proximal part of the fifth metatarsal were named 'tuberosity avulsion fracture' (Zone 1).

Still today, the most common classification of fractures to the proximal fifth metatarsal bone is into 'tuberosity avulsion fractures' (Zone 1), 'Jones fractures' (Zone 2) and 'diaphyseal stress fractures' (Zone 3) as shown in Figure 43.24a.

Fracture of the tuberosity

The fracture of the tuberosity at the base of the fifth metatarsal (Fig. 43.24b) has been described in Chapter 41 and is usually an avulsion injury that results from an acute ankle sprain. This uncomplicated fracture does well

treated with weight-bearing as tolerated with an elastic wrap circumferentially placed around the midfoot.¹¹⁰

Jones fracture

The fifth metatarsal fracture at the level of the articulation of the fourth and fifth metatarsals (Zone 2) is the Jones fracture (Fig. 43.24c). A Jones fracture may occur as the result of an acute plantarflexion/inversion mechanism or a medially directed force on a planted foot, and thus create an acute fracture. There may be a history of prodromal symptoms suggesting that the fracture may occur in an area already suffering bone stress. In Ekstrand's UEFA study of professional footballers with fifth metatarsal fractures, 45% reported pain on the lateral side of the foot prior to the acute fracture.¹¹¹

In 1984, Torg proposed managing these fractures on the basis of a radiographic (Fig. 43.24c) classification into three types.¹¹²

- Type 1: acute fractures characterised by a narrow fracture line and absence of intramedullary sclerosis.
- Type 2: those with delayed union, with widening of the fracture line and evidence of intramedullary sclerosis.
- Type 3: those with non-union and complete obliteration of the medullary canal by sclerotic bone.

Jones fractures can be treated operatively or non-operatively. In the 1990s, most Jones fractures were managed non-operatively with 6–8 weeks of non-weight-bearing.

However, there is now compelling evidence for screw fixation (Fig. 43.24d) of Jones fractures in sports medicine. Superior outcomes include both faster return to play (as early as 2–4 weeks) and fewer recurrences compared with cast treatment.^{113–115} The average time to return to sport following primary screw fixation is 7.5 weeks.¹¹⁶

Non-union may occur with this type of fracture because of the location of the nutrient artery (Fig. 43.24e).¹¹⁵

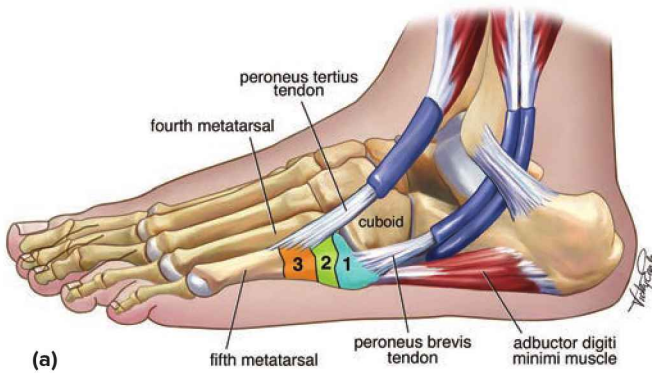
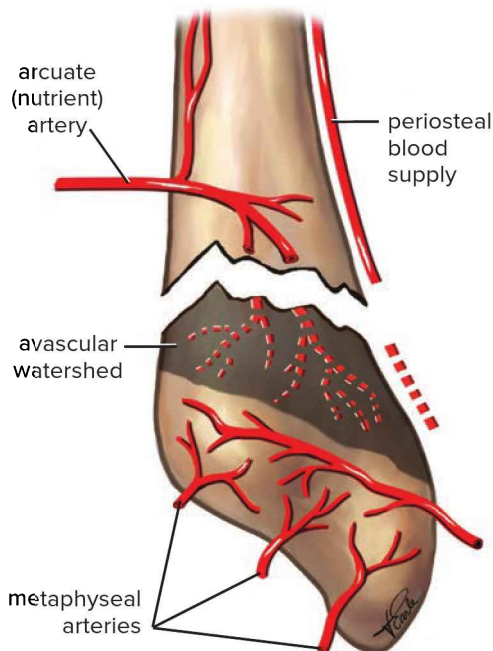


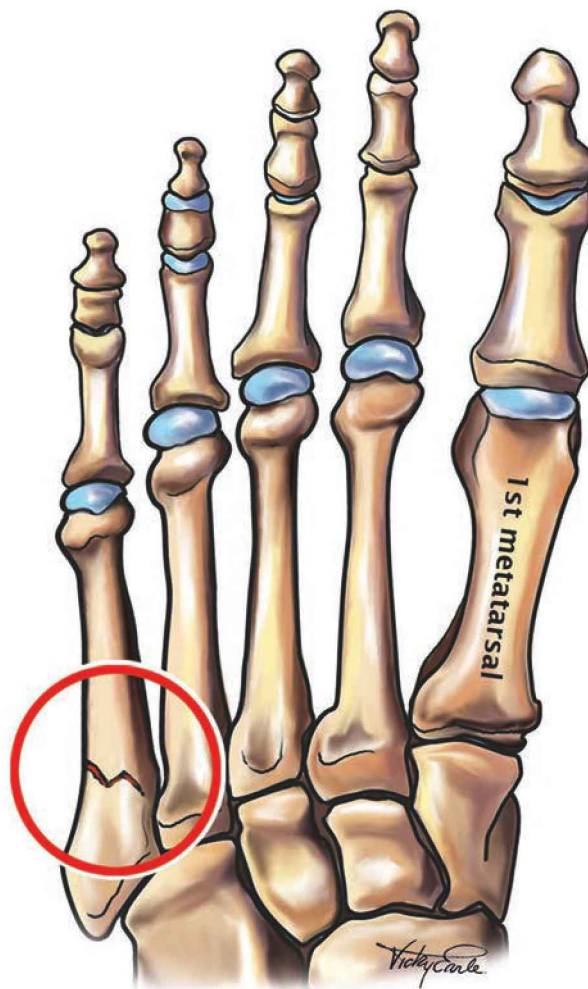
Figure 43.24 (a) Anatomical locations of the different types of proximal fifth metatarsal fractures. Zone 1 tuberosity avulsion fractures, Zone 2 Jones fractures and Zone 3 diaphyseal stress fractures (b) Fracture of the tuberosity of the fifth metatarsal (c) Jones fracture at the level of the fourth–fifth metatarsal articulation



(d)



(e)



(f)

Figure 43.24 (cont.) (d) Treatment with intramedullary screw (e) Arterial supply to the base of the fifth metatarsal (f) Stress fracture diaphysis of fifth metatarsal (g) Radiographic appearance of stress fracture diaphysis of fifth metatarsal

(e) FROM DEN HARTOG,¹⁶¹ ADAPTED FROM DAMERON TB JR. FRACTURE OF THE PROXIMAL FIFTH METATARSAL: SELECTING THE BEST TREATMENT OPTION. J AM ACAD ORTHOP SURG 1995;3:110-14

Re-fractures as a result of screw failure are fortunately uncommon, but are most likely to occur soon after return to full unrestricted sport. Re-fractures and non-unions are best treated with a large, 5.5 mm or larger, solid screw and bone grafting.¹¹⁷ The average time to return to sport following revision is 12 weeks. Early return to sport may predispose the athlete to re-fracture¹¹⁸ and it may be wise to wait for full radiographic healing before return to sport.¹¹⁹



(g)

Stress fracture of the diaphysis

Diaphyseal stress fractures occur distal to the intermetatarsal articulation (Zone 3) in the diaphysis of the bone (Fig. 43.24f and g). Similarly to Jones fractures, these fractures are prone to non-union and treatment with screw fixation (Fig. 43.24d) has been found to both reduce the incidence of non-union and result in improved return-to-sport times (6.9 weeks compared to 14.5 weeks in a non-operative group).¹¹⁴

Spiral fracture (distal third)

An acute spiral fracture of the distal third of the fifth metatarsal is seen, especially in dancers who suffer this fracture when they lose their balance while on demi pointe and roll over the outer border of the foot—the ‘fouetté fracture’ (Fig. 43.25). Non-displaced fractures may be treated with a plastic cast shoe, short leg walking boot or firm-soled shoe with weight-bearing as tolerated. Fractures that are displaced more than 3–4 mm or angulated more than 10° in the sagittal plane should be reduced.¹¹⁰

Approximately 25% of individuals who sustain a fifth metatarsal fracture—of any kind—continue to have some non-limiting pain up to one year later. Despite this low-level pain there were no functional impairments in a cohort of patients prospectively followed for one year.¹²⁰

MTP joint synovitis

MTP joint synovitis (commonly referred to as ‘metatarsalgia’) is a common painful condition occurring most frequently in the second, third and/or fourth MTP joints, or isolated in the first MTP joints. Tenosynovitis of the flexor tendon sheath is a potential differential diagnosis.¹²¹

The synovium of the MTP joints becomes painful, usually due to excessive pressure over a prolonged period. It is often related to:

- pes cavus or high-arched foot
- excessive pronation of the foot
- clawing or hammer toes
- tight extensor tendons of the toes
- prominent metatarsal heads
- Morton’s foot (Fig. 43.21): there is a shortened first metatarsal, which results in weight going through the second MTP joint.

The patient complains of pain aggravated by forefoot weight-bearing, particularly in the midstance and propulsion phases of walking. Pain onset is usually gradual.

Examination reveals local tenderness on palpation. Pain is aggravated by passive forced flexion of the toe. The V sign is sometimes seen in the early stages of metatarsal



Figure 43.25 Spiral fracture of the distal fifth metatarsal

joint synovitis. A separation of the toes can create a V shape that may indicate underlying dysfunction.¹²²

There may be an associated skin lesion (e.g. a callus) over the plantar surface of the affected joint due to the excessive load. This injury may be caused by uneven distribution of load, especially with excessive pronation.

Treatment involves addressing the most likely biomechanical contributing factors, as well as providing

symptomatic relief. Padding to redistribute weight from the painful areas can provide short-term relief. Footwear that has adequate midsole cushioning is essential. NSAIDs may be helpful and corticosteroid injection is occasionally required. The joint is injected via the dorsal surface, while longitudinal traction is placed on the toe to open the joint space.

First MTP joint sprain ('turf toe')

A sprain to the first MTP, otherwise known as a 'turf toe', is a common injury occurring in athletes in which the plantar capsule and ligament of the first MTP joint is damaged. The classic mechanism of injury is usually that of a forced hyperextension to the first MTP joint, although occasionally a plantarflexion injury to the joint may result in this injury.

Turf toe injuries are classified on the basis of their severity into grade I (mild), grade II (partial plantar capsule and ligament disruption) and grade III (complete plantar capsule and ligament disruption) (Table 43.6). Grade III injuries are often associated with injury of other structures such as the plantar plate or the flexor hallucis longus/flexor hallucis brevis tendons.¹²³

Predisposing risk factors include:

- competing or training on artificial turf
- pes planus or excessive pronation
- decreased pre-injury ankle range of motion
- decreased pre-injury MTP range of motion
- soft flexible footwear.

The athlete usually complains of localised pain, swelling and occasional redness at the first MTP joint following a 'bending' injury to the joint. The pain is classically aggravated by weight-bearing or movement of the big toe.

Examination reveals localised swelling and tenderness at the first MTP joint. In mild injuries plantar or plantar medial tenderness is present; in more severe injuries, dorsal tenderness occurs. Passive plantarflexion and dorsiflexion of the first MTP joint are generally painful and there may be a reduction in the range of movement in both directions. Passive gliding of the joint in a dorsal to plantar direction can be used to evaluate the plantar plate. Resisted plantarflexion and dorsiflexion can be used to evaluate the integrity of the flexor and extensor tendons.¹²³

Plain radiographs are generally unremarkable, although occasionally small periarticular flecks of bone are noted, most likely indicating avulsion of the MTP capsule or ligamentous complex. If injury to the plantar plate is suspected, lateral views in forced dorsiflexion can be used. This view may demonstrate a lack of distal sesamoid excursion, which is indicative of complete disruption of the plantar ligament structures.¹²³ MRI may be appropriate for elite athletes with grade II and III injuries to outline the extent of ligamentous, osseous and cartilage damage.¹²⁴

Treatment consists of ice, NSAIDs, electrotherapeutic modalities and decreased weight-bearing for at least 72 hours. Additional treatment may include taping (Fig. 43.26) and the use of stiff-soled shoes or a CAM walker boot to protect the first MTP joint from further injury. Treatment of the various grades of injury is summarised in Table 43.6. Recovery from grade I and II turf toe injury generally takes 3–4 weeks. Surgical indications for grade III injuries include:¹²³

- large capsular avulsion and vertical instability
- retraction of sesamoids
- loose body
- chondral injury.

A possible long-term sequel to this injury is the development of hallux limitus, discussed in the next section.

Table 43.6 Turf toe grading and treatment

Grade	Description/Findings	Treatment	Return to play
I	Attenuation of plantar structures Localised swelling Minimal ecchymosis	Symptomatic	Return as tolerated
II	Partial tear of plantar structures Moderate swelling Restricted motion due to pain	Walking boot Crutches as needed	Up to 2 weeks May need taping on return to play
III	Complete disruption of plantar structures Significant swelling/ecchymosis Hallux flexion weakness Frank instability of hallux MTP	Long-term immobilisation in boot or cast OR Surgical reconstruction	6–10 weeks depending on sport and position Likely need taping on return to play

ADAPTED FROM MCCORMICK AND ANDERSON¹²³ REPRODUCED WITH PERMISSION OF JOURNAL OF CLINICAL SPORTS MEDICINE



Figure 43.26 Taping to protect an injured first MTP joint

Hallux limitus

Hallux limitus refers to restriction in dorsiflexion of the hallux at the first MTP joint secondary to exostoses or osteoarthritis of the joint. Often the term 'hallux rigidus' is used to describe the final progression of hallux limitus as ankylosis of the joint occurs.

The primary role of the hallux is to enable dorsiflexion of the first metatarsal during the propulsive phase of gait. Approximately 60° of dorsiflexion is required for normal gait. Limitation of this range of motion results in problems with gait. Normal dorsiflexion is achieved via the action of peroneus longus, which helps to stabilise and maintain plantarflexion of the first ray.¹²⁵

There is some evidence that the following may contribute to development of hallux limitus:

- trauma: secondary to chondral damage
- excessive load: via absolute loading or due to excessive pronation of the foot which increases the stresses on the joint and promotes development of exostoses
- repetitive weight-bearing dorsiflexion of the first MTP joint
- autoimmune arthropathy or metabolic conditions (e.g. rheumatoid arthritis, psoriatic arthritis, ankylosing spondylitis, gout)¹²⁵
- hypermobile first ray (e.g. reduced plantarflexion of the first ray can increase compressive forces in the joint during propulsion).¹²⁵

The main symptom is usually pain around the first MTP joint—often described as a deep aching sensation. Pain is aggravated by walking, especially in high heels, or activities involving forefoot weight-bearing. Dorsal joint hypertrophy can cause footwear to impinge on the joint and may lead

to pain secondary to skin or soft tissue irritation. In patients with longstanding hallux limitus, a distinct shoe wear pattern is seen: the sole demonstrates wear beneath the second MTP joint and the first interphalangeal joint.

Examination reveals tenderness over the dorsal aspect of the first MTP joint, often with palpable dorsal exostoses. There is a painful limitation of joint motion. Some with this condition may develop chronic subungual haematoma due to repetitive compensatory dorsiflexion in the interphalangeal joint.¹²⁵

Plain radiographs display the classic characteristics of degenerative osteoarthritis and the degree of degeneration. Features include joint space narrowing, sclerosis of the subchondral bone plate, osteophytic proliferation, flattening of the joint, sesamoid displacement and free bony fragments.

Conservative management consists of an initial reduction in activity, NSAIDs, a corticosteroid injection as adjuvant therapy, passive joint mobilisation before the joint fully fuses, correction of abnormal biomechanics with orthoses and/or footwear, and dynamic splinting.¹²⁶ Conservative treatment often fails when hallux dorsiflexion is less than 50°.

If conservative treatment fails, cheilectomy (resection of all bony prominences of the metatarsal head and base of the proximal phalanx) is required. Occasionally, arthroplasty of the first MTP joint is indicated.

Hallux valgus

Hallux valgus refers to static subluxation of the first MTP joint. It is characterised by valgus (lateral) deviation of the great toe and varus (medial) deviation of the first metatarsal (Fig. 43.27). Bony exostoses develop around the first MTP joint, often with an overlying bursitis. In severe cases, exostoses limit first MTP joint range of motion and cause pain with the pressure of footwear. Hallux valgus is common, affecting 23% of adults aged between 18 and 65 years.¹²⁷ It is more common in women. Although it is more common in older athletes, it can also develop in adolescence.¹²⁸

The development of hallux valgus appears to occur secondary to a combination of intrinsic (genetic) and extrinsic causes. There is some evidence for an association of the condition with:

- constricting footwear (e.g. high heels)
- excessive pronation: increased pressure on the medial border of the hallux, resulting in deformation of the medial capsular structures
- increased length of the first metatarsal and hallux¹²⁹
- trauma to the medial and plantar ligament complex and medial sesamoid bone¹²³
- others including cystic degeneration of the medial capsule, Achilles tendon contracture, neuromuscular disorders and collagen-deficient diseases.



Figure 43.27 Hallux valgus

In the early phases, hallux valgus is often asymptomatic but as the deformity develops, pain arises at the medial eminence. The pain is typically relieved by removing shoes or by wearing soft, flexible, wide-toed shoes. Blistering of the skin or development of an inflamed bursa over the medial eminence may occur. In severe deformity, lateral metatarsalgia may occur due to the diminished weight-bearing capacity of the first ray.

Examination reveals the hallux valgus deformity often with a tender swelling overlying the medial eminence. Plain radiographs can image both the severity of the deformity and the degree of first MTP joint degeneration.

Initial treatment involves appropriate padding and footwear to reduce friction over the medial eminence. Correction of foot function with orthoses is essential. In more severe cases, surgery may be required to reconstruct the first MTP joint and remove the bony exostoses.

Minimally invasive surgical techniques for hallux valgus have been developed.¹³⁰ Orthoses are often required after surgery and gait re-education is needed to restore normal weight-bearing through the first ray.¹³¹

Sesamoid injuries

The first MTP joint consists of the metatarsophalangeal joint and two regularly occurring sesamoid bones, that play a significant role in great toe function. Embedded within the two tendons of the flexor hallucis brevis, the sesamoids:

- functionally lengthen the first metatarsal
- protect the tendon of flexor hallucis longus
- transfer weight-bearing from the lateral to the medial aspect of the forefoot.

Bipartite sesamoids occur in approximately 30% of individuals and are bilateral 25% of the time.

The sesamoid bones can suffer stress fracture, sesamoiditis ('chondromalacia'), acute fracture, osteochondritis, osteoarthritis and bursitis.¹³² Sesamoid stress fractures may occur in many sporting activities, but particularly in basketball players, tennis players and dancers. Most sesamoid abnormalities involve inflammatory changes and osteonecrosis. The medial sesamoid is more frequently affected. Injury may be caused by landing after a jump, increased forefoot weight-bearing activities (e.g. sprinting and dancing) or after traumatic dorsiflexion of the hallux. Pronation and progressive hallux valgus positioning may cause lateral displacement or subluxation of the sesamoids within the plantar grooves of the first metatarsal. This subluxation of the sesamoids may lead to erosion of the plantar aspect of the first metatarsal, resulting in pain underneath the first metatarsal head and arthritic changes reminiscent of patellofemoral pathology.

The patient complains of pain with forefoot weight-bearing and may display a supinated antalgic gait to avoid painful toe-off. Examination reveals marked local tenderness and swelling overlying the medial or lateral sesamoid. Movement of the first MTP joint is usually painful and often restricted. Resisted plantar flexion of the great toe is weak and painful.

Plain radiographs for this bone require an anteroposterior (AP), lateral and axial sesamoid view but radiographs are often not diagnostic for stress injury. They may show an acute fracture or bony sclerosis of sesamoid necrosis. MRI is often required to detect early stress fractures or more specific pathologies as mentioned above.

Recent onset or mild cases of sesamoid pain are traditionally treated with load modification, ice and possibly NSAIDs. Appropriate accommodative padding can partially relieve direct pressure to the sesamoids (Fig. 43.28). Orthoses can include this accommodation within the cushioning layer of the device. A carefully placed corticosteroid injection may provide pain relief for those athletes who must delay a rest period but it is not recommended. Ideal initial treatment of sesamoid



Figure 43.28 Sesamoiditis—padding

stress fractures includes a several-week period of non weight-bearing in a boot or short leg cast. The duration is determined by frequent examination awaiting resolution of tenderness by firm palpation.

Gradual return to full weight-bearing can be bridged by using a highly curved stiff shoe or rocker bottom shoe. Transition to athletic activities should also be assisted

with the previously discussed orthosis which unloads the first MTP and places the foot into a laterally directed supinated position thereby reducing the toe-off force through the first MTP.

Non-operative management is preferable, but a few cases will fail this approach. Surgical fixation or sesamoidectomy is reserved for these recalcitrant few. Sesamoidectomy does risk developing hallux deformities. A recent surgical series estimated approximately 12 weeks to return to sport.¹³³

Plantar plate tear

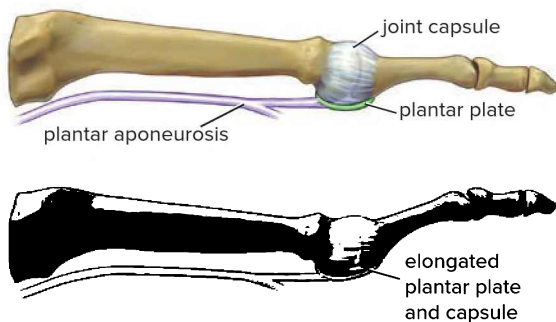
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Tears of the plantar plate may be the most common cause of pain under the second MTP joint, though it can occur at any of the MTP joints. It is also referred to as pre-dislocation syndrome, crossover toe deformity and floating toe syndrome. A fibrocartilaginous thickening of the MTP joint (Fig. 43.29a), the plantar plate:

- stabilises the MTP joint
- assists in the windlass mechanism due to its attachment to the plantar fascia
- resists hyperextension of the MTP joint
- absorbs compressive loads.

The second MTP joint is most likely to be affected as it is the longest metatarsal and has unopposed lumbricals and no plantar interossei insertions. Plantar plate tears usually result from repetitive overload from abnormal forefoot loading patterns resulting from hallux valgus, excessive pronation, short first metatarsal or long second metatarsal. The tear usually arises from the base of the proximal phalanx.¹³⁴

The athlete usually complains of localised pain under the MTP joint. Swelling may be present, extending to the dorsal aspect of the joint. Neuroma-like symptoms may be experienced by patients due to irritation of the plantar digital nerve from the localised oedema.



(a)

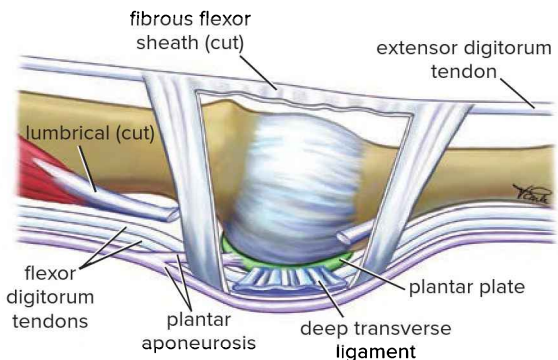


Figure 43.29 (a) Anatomy of the MTP joint demonstrating the plantar plate



(b)

Figure 43.29 (cont.) (b) Modified Lachman's anterior-posterior drawer test. The metatarsal head is stabilised and the proximal phalanx is dorsally translocated. A 2 mm or 50% joint displacement is a positive sign of plantar plate laxity (c) Taping. Plantarflexion strapping of the digit

Examination reveals pain at the base of the proximal phalanx which may be aggravated by dorsiflexion of the joint. In relaxed stance, a dorsiflexion deformity of the toe may be noted. This is often accompanied by a crossover deformity. The modified Lachman's test (anterior-posterior drawer) can also be utilised. The metatarsal head is stabilised and the proximal phalanx is dorsally translocated (Fig 43.29b). A 2 mm or 50% joint displacement is a positive sign of plantar plate laxity.

Ultrasound may reveal a hypoechoic defect in the plantar plate, usually at the distal attachment. Historically arthrography was used to demonstrate synovial hypertrophy and extravasation of dye into the flexor tendon sheath. Today, MRI is helpful to demonstrate a tear of the plantar plate. There is increased signal intensity within the plate, along with a loss of continuity.¹³⁵

Treatment initially consists of ice, NSAIDs, relative rest, plantarflexion strapping of the digit (Fig 43.29c) and accommodative padding to reduce loads under the affected MTP joint. Orthoses and a stiff-soled shoe or rocker bottom shoe are also required. Some clinicians consider an extra-articular corticosteroid injection may also be useful but there have been no quality trials of this treatment. Primary repair of the plantar plate with or without a flexor tendon transfer may be required in patients that fail to respond to conservative measures.



(c)

Corns and calluses

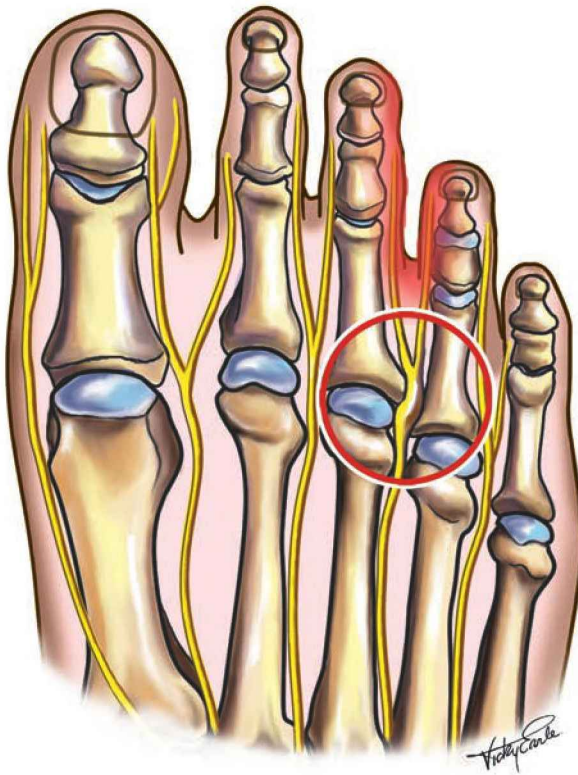
Excessive pressure on the skin may cause hypertrophy of the squamous cell layer of the epidermis, which manifests as corns and calluses. In the feet, corns and calluses result from uneven weight distribution and thus indicate abnormal foot biomechanics or poorly fitting footwear.

Treatment involves the removal of circumscribed corns and diffuse areas of callus with a scalpel, the wearing of well-fitting footwear and, if abnormal foot mechanics are present, orthoses. Petroleum jelly over the corn or callus and on the outside of the sock can also help.

Morton's interdigital neuroma

So-called Morton's neuroma is not a true neuroma, but a swelling of nerve and scar tissue arising from compression of the interdigital nerve, usually between the third and fourth metatarsals (Fig. 43.30a). Excessive pronation contributes to metatarsal hypermobility and impingement of the interdigital nerve.

The patient complains of pain radiating into the toes, often associated with pins and needles, and numbness. A clicking sensation may be noticed. Pain is increased by forefoot weight-bearing activities and with narrow-fitting footwear, and is often relieved by removing the shoes.



(a)



(b)

Figure 43.30 Morton's neuroma (a) Location of nerve entrapment in Morton's neuroma (b) Plantar metatarsal padding

Examination reveals localised tenderness and, in cases of extensive chronic proliferation, there may be a palpable click on compression of the metatarsal heads. Web space tenderness and toe tip sensation deficit may be present.¹³⁶

Diagnostic confirmation with imaging can be performed for chronic cases considering surgical management. MRI demonstrating a neuroma greater than 5 mm is generally significant. Ultrasound has emerged as a viable technique to diagnose and distinguish an interdigital nerve from an intermetatarsal bursa.^{137, 138}

Treatment consists initially of ice to alleviate acute tenderness. Plantar metatarsal padding to subtly separate the metatarsals and shoes with a wide toe box are used to reduce the side-to-side compression of the metatarsals (Fig. 43.30b). However, in chronic cases, little improvement is seen with padding. Intrinsic and extrinsic foot muscle strengthening exercises are indicated to maintain or improve the transverse arch and control of foot pronation. The use of orthoses is essential if excessive pronation is present.

Various interventions have been proposed.¹³⁹ Injection of corticosteroid and local anaesthetic agents in conjunction with the padding may provide lasting relief. An ultrasound-guided corticosteroid injection via a distal interdigital approach provided at least 9 months significant relief of symptoms for the majority of patients.¹⁴⁰ Injection with alcohol is also an option.^{141, 142}

If the patient obtains no relief, radiofrequency ablation of the nerve appears to be a promising new therapy.¹⁴³ Recently, sonographically guided cryoneurolysis improved four of five Morton's neuroma patients.¹⁴⁴

Surgical management is quite successful for those cases not responding to conservative care. Pain relief remains the principal goal of treatment; however, a resected neuroma will result in permanent numbness and this needs to be discussed with the patient prior to surgery. Decompression by neurolysis has been described as a successful intraoperative decision to preserve the nerve in some cases.¹⁴⁵

Plantar warts

The papovavirus causes plantar warts when it enters the skin. The warts can be particularly painful on weight-bearing. They should be differentiated from corns. Applying lateral pressure or pinching will be painful in warts, whereas corns are more painful with direct pressure. Gentle paring with a scalpel will also reveal the characteristic appearance of a plantar wart with fine black dots within a defined margin of white or brown tissue.

Plantar warts are best treated with chemical solutions containing salicylic acid. The overlying hyperkeratosis should be removed weekly to allow the chemicals to penetrate the wart. Blistering and abscess formation occur and require debridement with a scalpel and the application of a dressing.

Onychocryptosis

Onychocryptosis (ingrown toenail) arises from abnormal nail growth or poor nail cutting. Patients often present in acute pain with tenderness on gentle palpation. Nails are often infected. Treatment with local and oral antibiotic therapy is required. Conservative treatment involves elevating the offending corner of the nail plate from the soft tissue, cutting a small 'V' into the middle of the nail distally (to take pressure off the edges) and stretching the soft tissue away from the nail with a cotton bud.

Surgical management consists of resection of the outer aspect of the nail to prevent the nail border injuring the soft tissue. Resection involves anaesthetising the toe and removing the nail border with appropriate nail splitters and forceps. All abnormal tissue is removed and the nail matrix treated with phenol to prevent regrowth.

Less common causes of forefoot pain

Stress fracture of the great toe

Stress fractures of the proximal phalanx of the great toe have been reported in adolescent athletes.^{99, 146} Stress fractures in the distal phalanx can occur in ballet dancers. There appears to be an association with hallux valgus. Treatment involves a period of non weight-bearing rest of 4–6 weeks followed by graduated return to activity.

Joplin's neuritis

Joplin's neuritis involves compression and irritation of the dorsal medial cutaneous nerve over the first metatarsal and first MTP joint. It usually occurs because of irritation from footwear and is common in patients with hallux abducto-valgus or exostoses around the first MTP joint.

The patient complains of pain radiating along the first ray into the hallux. Wearing appropriate footwear and using foam and felt to redistribute the load from the affected area generally provides relief. Orthoses may be required to prevent excessive pronation.

Freiberg's osteochondritis

Freiberg's disease or osteochondritis of the metatarsal head affects adolescents between the ages of 14 and 18 years (Chapter 44). The metatarsal head appears fragmented on radiograph. Offloading of the metatarsal heads using padding and orthoses is essential to prevent permanent metatarsal head flattening that may predispose to adult osteoarthritis.

Toe clawing

Toe clawing occurs secondary to short, tight long flexor tendons (Fig. 43.31). During the propulsive phase of gait, the long flexors contract to stabilise the toes. In the unstable foot, these long flexors contract excessively during the propulsive phase and the toes claw the surface in an attempt to maintain stability.

Clawing of toes does not result in pain itself but excessive pressure on the prominent joints and ends of toes often causes painful skin lesions.⁸⁸

Subungual haematoma

Subungual haematoma occurs when direct trauma or repetitive pressure from footwear leads to bleeding under the toenails. Pain arises from increased pressure under the nail and the nails appear black. The nail may eventually loosen from the nail bed.

Treatment of an acute subungual haematoma involves using a heated needle or paperclip to perforate the nail and release the collection of blood. Cover with a small dressing. Correctly fitting footwear and appropriate lacing techniques may prevent recurrence.

Subungual exostosis

A subungual exostosis develops because of direct trauma. The patient presents with pain on direct pressure to the nail. The nail plate may be displaced from the nail bed due to elevation from the exostosis. Treatment requires the wearing of loose-fitting footwear, appropriate cutting of the nail and padding. Surgery may be required to remove the bony exostosis.

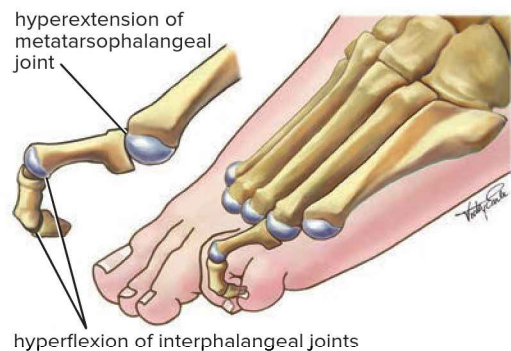


Figure 43.31 Toe clawing

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e